Mapping Emerging Digital Technologies in Defense: A Scientometric and Systematic Review

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***Abstract*—The current paper presents a scientometric and systematic analysis of defense evolution driven by emerging technologies, focusing on two broad categories: 1) internal defense (civil) and 2) external defense (ground, aerial, and naval). The study explores how technologies such as artificial intelligence (AI), machine learning (ML), Internet of Things (IoT), digital twin (DT), and next-generation networks (4G, 5G, and 6G) transform defense operations by enhancing efficiency, accuracy, and decision-making capabilities. The analysis integrates three tools: 1) excel for examining publication and citation trends; 2) CiteSpace for citation network visualisation and co-citation cluster identification; and 3) VOSViewer for keyword co-occurrence mapping and geographic contribution analysis. The dataset, comprising records retrieved from the Scopus database, spans the publication years 2019 to 2024. In addition, a systematic literature review (SLR) was conducted covering English-language peer-reviewed journal and conference articles published between 2016 and 2025. The findings reveal a significant rise in defense-related research, particularly from 2019 onward, driven by the integration of AI-enabled autonomous systems, predictive analytics, and real-time situational awareness. The study concludes with future research directions emphasising the need for 6G-enabled DTs, advanced AI capabilities, and cross-domain interoperability to enhance the adaptability, resilience, and intelligence of modern defense systems.**

***Index Terms*—Aerial defense, civil defense, ground defense, naval defense, scientometric analysis.**

I. Introduction

ARTIFICIAL intelligence (AI), machine learning (ML), the Internet of Things (IoT), and digital twin (DT) have expanded beyond traditional domains such as education, manufacturing, and healthcare, and now assume a crucial role in national security, particularly in defense. The emerging technologies are transforming defense strategies by enhancing situational awareness, optimising operations, enhancing security, and enabling autonomous decision-making [1], [2]. Historically, nations have pursued technological advancements to achieve military superiority, with new technologies consistently reshaping warfare by introducing advanced weapons and altering military tactics [3], [4]. Nowadays, AI enables real-time image and video processing for battlefield surveillance, object recognition, and target identification. Intelligence gathering is enhanced through satellite imagery, signal intelligence, and social media analysis, contributing to improved situational awareness and threat detection [5], [6].

A nation's defense is broadly categorised into two main types.

1) *Internal Defense:* Managed by law enforcement agencies such as the police or civil defense forces, ensuring domestic security and stability.
2) *External Defense:* Comprising ground (army), aerial (air force), and naval (navy) forces, responsible for safeguarding national sovereignty against external threats.

Both developing and developed nations are making substantial investments in AI, IoT, autonomous aerial vehicles (UAVs), and DT technologies (DTTs). In 2021, the national security commission on artificial intelligence (NSCAI) recommended a $40 billion investment in AI research, development, and innovation to maintain technological superiority [7]. Between FY2019 and FY2025, the U.S. Department of Defense significantly expanded AI funding across all warfighting domains, with over 25% allocated to multidomain applications, underscoring AI's strategic role in integrated defense modernization.[1]

1) *Ground Defense:* Ground defense, a critical component of external defense, is rapidly evolving with AI, ML, DT, and next-generation networks (5G/6G) technologies. The modern battlefield demands AI-driven solutions for superior decision making and operational effectiveness [8]. The U.S. Army's TRADOC prioritises AI-powered autonomous systems to enhance situational awareness, reduce cognitive load, and optimise operations. Global militaries are actively exploring 5G networks for low-latency communication, while 6G, still in development, promises ubiquitous connectivity and real-time data exchange [9]. DT technology is transforming ground defense by creating virtual replicas of vehicles, equipment, and infrastructure, enabling real-time simulation, predictive maintenance, and performance optimisation [10], [11]. DT enhances operational readiness by identifying vulnerabilities and reducing downtime. Key applications include autonomous ground vehicles (UGVs) for mobility, advanced surveillance for target recognition,

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[1]NSCAI, Final Report, 2021.

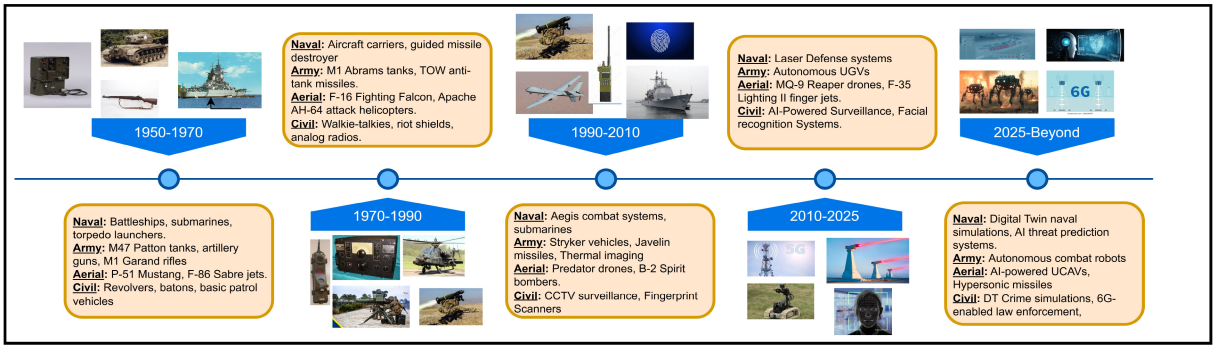

Fig. 1. Evolution of defense research from 1950 to beyond 2025.

human-machine teaming for rapid threat response, and DT-powered cybersecurity for electronic warfare [12], [13]. Generative AI and cyber–physical systems, when combined with the Internet of Robotic Things (IoRT) and DTT, enable smart, immersive environments for real-time simulation, autonomous decision-making, and collaborative robotics; when applied to ground defense, ML-based clustering and spatial cognition algorithms further enhance autonomous navigation, route optimization, and task execution in UGVs, thereby improving battlefield decision-making and operational efficiency [14], [15]. As AI, DT, and 5G/6G continue to advance, the integration will significantly strengthen military capabilities and national security [12].

2) *Naval Defense:* Modern navies increasingly leverage digital transformation to enhance operational effectiveness, safety, and decision-making. Research in defense and maritime technology highlights how integration of AI, ML, IoT, and DTT supported by ultrafast 5G/6G networks significantly improves real-time situational awareness, predictive maintenance, and autonomous system coordination. AI and ML algorithms analyse vast amounts of sensor data, reducing response times by up to 25% in threat-detection scenarios compared to traditional methods [16], [17]. Predictive models based on ML forecast impending system failures before their occurrence. IoT devices are increasingly deployed on ships and naval installations to monitor everything from engine performance to environmental conditions. IoT-enabled frameworks boost overall operational efficiency by approximately 20% through real-time tracking and early fault detection [18]. Additionally, 5G networks provide data rates up to 10 times faster than previous generations, with ultralow latency crucial for controlling autonomous systems and facilitating real-time data exchange. High-speed communication supports seamless integration of IoT sensors, AI analytics, and DT platforms even in challenging maritime conditions, ensuring critical information remains continuously available to decision-makers [19], [20].

3) *Aerial Defense:* The synergistic integration of AI/ML, IoT, DT, and advanced communication networks creates a robust, interlinked defense system. AI and ML techniques, applied to data from radar systems, satellite imagery, and onboard sensors, can identify and classify aerial threats with remarkable speed and precision. In parallel, UAVs have emerged as a critical asset in modern naval operations. In 2023, the autonomous UAV sector was estimated at U.S. \$15 634.7 million, with forecasts predicting growth to approximately U.S. \$91 304.8 million by 2033. By 2025, global economic benefits from autonomous vehicles are projected to reach U.S. \$200 billion to 1.9 trillion [21]. The market value for UAVs in agriculture is projected to reach U.S. \$5.19 billion by 2025 [22]. Recent advancements in UAV power systems, driven by the development of efficient batteries, have notably extended flight times. Moreover, high-resolution cameras and advanced sensors captured real-time data during flight, enabling rapid decision-making and enhancing situational awareness [23], [24].

4) *Civil Defense:* Advanced technologies are transforming internal defense systems. AI and ML algorithms analyse data from diverse sources such as CCTV feeds, social media, and sensor inputs to detect anomalies and potential threats. ML models process historical crime data along with real-time inputs to predict crime hotspots, enabling predictive policing, which optimises patrol routes and resource allocation, thereby reducing crime rates in high-risk areas. IoT sensor networks significantly boost situational awareness by providing instant access to environmental data during emergencies like fires, chemical spills, or terrorist attacks. Advanced communication networks further enhance the capability by facilitating rapid information dissemination and decision-making through real-time data transfers, which are crucial for coordinating effective responses [25], [26]. Fig. 1 outlines the chronological progression of defense research from 1950 to beyond 2025, highlighting domain-specific advancements across naval, aerial, ground, and civil sectors. The progression from conventional warfare systems to AI-driven, autonomous, and network-enabled technologies is illustrated. Fig. 2 illustrates the integration of emerging technologies, such as AI, ML, IoT, DT, and 4G/5G/6G across defense domains. Each technology enhances operational goals

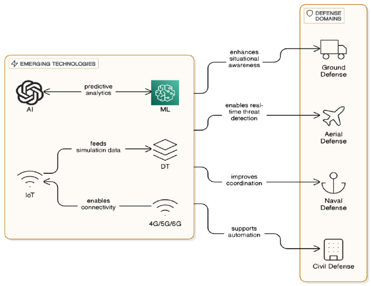

Fig. 2. Defense emerging technologies integration.

such as situational awareness, real-time threat detection, coordination, and autonomous surveillance, reinforcing a shift toward intelligent and connected defense infrastructures. Generative AI and DTT are increasingly extending beyond defense to transform productivity, decision-making, and operational innovation across industrial and financial ecosystems [27].

A. Multilayered Architecture for Defense Data Processing and Decision Making

The section details the step-by-step process of communication and decision-making in defense operations. Fig. 3 illustrates a real-time data processing framework where information flows from the Data Collection Layer through communication, processing, and AI/ML decision layers to support timely responses. Each layer refines data and sends feedback to enhance accuracy, enabling adaptive, intelligent decision-making across defense operations.

1) The *Data collection layer* forms the foundation of the defense architecture, gathering real-time data from various sources such as sensors, UAVs, maritime systems, and more. The Data Collection Layer transmits the collected data to the Communication Layer for secure delivery to higher layers. The layer operates in a bidirectional flow, receiving control signals and configuration updates from higher layers. The updates influence how and what data is collected, such as adjusting sensor sensitivity, modifying UAV routes, or changing data collection frequency [28].
2) The *Communication layer* serves as the central highway, ensuring fast, secure, and reliable data transmission across the system. Communication Layer sends data from the Collection Layer to the Processing Layer while also receiving feedback and control signals from higher layers. To maintain data integrity, the Communication Layer detects and corrects communication errors, sending retransmission requests if data packets are lost or corrupted. The bidirectional flow enables continuous feedback, improving accuracy and responsiveness in defense operations [29], [30].
3) The *Processing layer* functions as the core data refinement and transformation hub. The Layer receives raw inputs such as sensor feeds, UAV telemetry, and maritime diagnostics from the Communication Layer. The layer performs essential preprocessing tasks, including noise reduction, de-duplication, data normalisation, and contextual tagging to ensure semantic consistency across multisource datasets. The Layer prepares the data for accurate interpretation by AI/ML models in the next layer. Critically, the Processing Layer features a dynamic rule-based engine which updates in real time based on feedback from the AI/ML Decision Layer. For instance, if AI models identify recurring false anomaly detections due to environmental noise (e.g., ocean wave interference in naval systems), the Processing Layer refines filtering thresholds to eliminate irrelevant signals. The bidirectional learning loop ensures the system continuously evolves, improving accuracy, scalability, and context-aware adaptability, especially vital in fast-paced defense environments where data reliability directly impacts decision-making.

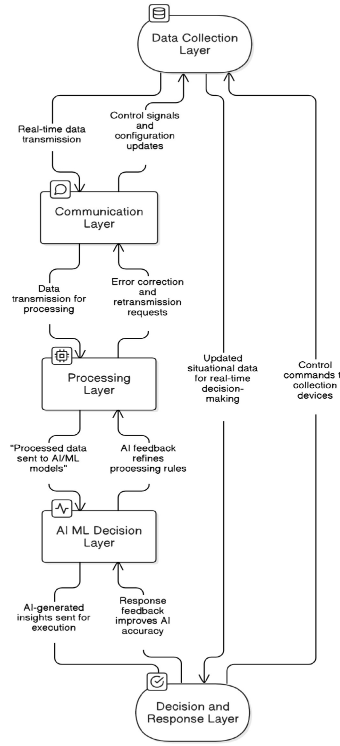

Fig. 3. Multilayered data flow framework for defense communication and decision-making.

4) The *AI/ML decision layer* plays a vital role in transforming refined data into actionable insights using AI and ML algorithms. The AI/ML Decision Layer handles data analysis, pattern recognition, predictive modelling, and decision-making, automating and optimising defense operations. The layer receives refined data from the Processing Layer and applies AI models to detect anomalies, such as unauthorised access or suspicious movements. ML algorithms identify unusual patterns by comparing real-time data with historical datasets and predict system failures through equipment health analysis. AI/ML-driven decision-making powers autonomous defense systems, including UAV navigation, robotic patrols, and fleet optimisation. The layer maintains a bidirectional connection with the Processing

Layer, sending feedback signals to refine data filtering and structuring algorithms based on AI insights. The AI/ML Decision Layer transmits actionable insights to the Decision and Response Layer, including threat alerts, reports, and automated system commands, enhancing real-time defense decision-making and operational efficiency [31].

5) The *Decision and response layer* translates insights from the AI/ML Decision Layer into actionable outcomes. The Layer enables automated response execution (such as UAV route correction, activation of naval sensors, or system lockdown protocols) while supporting human-in-the-loop oversight for high-risk or strategic operations (e.g., airstrike authorisations or border control alerts). What makes the layer critical is its closed-loop feedback system. Post-action metrics such as mission outcome success, false alarm rate, and operator overrides are analysed and sent back to the AI/ML and Processing Layers to fine-tune thresholds, retrain models, and adjust data collection strategies. For example, if a surveillance drone is frequently redirected without cause, the system can infer over-sensitivity and recalibrate its anomaly detection logic. The performance-aware learning cycle ensures the system not only reacts but also improves over time, aligning machine decisions with mission priorities and operational realities.

Paper Organisation: Section II discusses the related work in the current study domain. Section III presents the detailed research methodology. Section IV depicts the detailed scientometric analysis. Section V presents the detailed systematic literature review (SLR), and Section VI depicts the discussion and possible future research directions. Finally, Section VII concludes this article.

II. Related Work

The section focuses on the existing survey studies in defense all categories, such as ground, aerial, naval, and civil. Kaur and Saini [32] identified five AI-based clusters, reflecting emerging research trends and AI methodologies in crime prediction. The scientometric analysis of crime prediction and pattern analysis research over the past decade, based on Scopus data, emphasises citation trends, prominent journals, key institutions, and author collaborations. Rodriguez et al. [33] conducted a scientometric and bibliometric analysis on UAV applications in architecture and urbanism, using VOSViewer to identify trends, techniques, and fields of application such as 3-D modelling, land mapping, and construction monitoring. The study highlights collaboration gaps between prolific authors across continents and emphasises future trends, including UAV integration with IoT, 5G, AI, and ML for enhanced real time data collection and analysis. Wang et al. [34] highlighted the UAV remote sensing offers significant advantages over traditional satellite, including higher resolution, affordability, and faster data acquisition, leading to widespread adoption across disciplines. The analysis reveals, China and the U.S. lead in publications, with future trends pointing toward AI-driven applications. Yang et al. [35] investigated the growing research interest in autonomous surface vessels (USVs) for industry, military, and ocean engineering, highlighting recent technological advancements. Through a bibliometric analysis (2000–2023), the study identifies key research hotspots, trends, and modeling techniques. Six future directions are proposed: 1) enhanced autonomy; 2) integrated sensor systems; 3) extended endurance; 4) satellite communication; 5) eco-friendly practices; and 6) improved defense capabilities, while stressing the need for industry-academia collaboration to drive innovation. Pandey et al. [36] analysed 417 articles (1991–2023) on AI and ML in defense, highlighted a post-2017 surge, 2.49% annual growth, and 14.63% international collaboration. The USA, China, India, U.K., and Korea lead in publications, with the USA most cited. The study emphasizes AI-driven automation, cybersecurity, and strategic decision making, while addressing ethical and legal concerns of autonomous systems. Taranovskiy and Yashnikova [37] analysed scientific periodicals on marine navigation equipment, identifying top global and Russian journals. The study highlights the need for improved editorial policies, broader author collaboration, and enhanced content quality, contributing to advancements in naval and defense navigation technology. In aerial defense, Keneni et al. [38] presented an XAI model for UAVs to explain mission decisions. By using Sugeno-type fuzzy inference systems (FISs) with simulated mission data, the model predicts the impact of weather and enemy locations when UAVs deviate from the path. The process consists of two phases, combining Mamdani FIS for rule creation and Sugeno FIS with subtractive clustering for reverse modeling, which achieves high accuracy with the root mean square error (RMSE) below 0.05, making it highly relevant to UAV navigation and threat response in aerial defense operations. Alquwayzani and Albuali [39] conducted a systematic review on UAV security, highlighting the cyber risks associated with the growing use in defense, logistics, and surveillance. The study emphasises the Zero Trust architecture as an effective solution to mitigate cyber threats, reducing vulnerabilities by enforcing continuous verification and monitoring of UAV systems. The integration of ML is discussed for enhancing threat detection and preventing unauthorised access, positioning Zero Trust as a robust approach to UAV cybersecurity. Existing literature across defense categories provides valuable insights, yet several gaps remain unresolved, such as the absence of cross-domain scientometric analysis, fragmented evaluations of emerging technologies, limited focus on layered defense architectures, and insufficient attention to future research trajectories. The current paper aims to bridge the gaps by presenting a comprehensive cross-domain scientometric review, analysing technological convergence, introducing a layered defense architecture framework, and offering insights into global collaboration patterns along with emerging trends and future directions. A DT-based synthetic training framework is adopted to address the lack of annotated data and user behavior bias in defense surveillance by simulating realistic object interactions using 3-D models [40]. For instance, the creation of Eva Herzigová's lifelike DT using motion capture and extended reality technologies highlights how immersive technologies are redefining human presence and interaction in

TABLE I
RESEARCH STUDIES ON DEFENSE AND SECURITY

Reference	Approach	Visualization Tool	CoNA	CcNA	Research Challenges	Domain
[32]	Scientometric	VOSviewer	✓	✓	✓	Civil
[35]	Bibliometric	CiteSpace	✓	✓	✓	Naval
[36]	Bibliometric	CiteSpace	✓	✓	✓	Ground
[37]	Scientometric	✗	✓	✓	✗	Naval
[39]	SLR	✗	✓	✓	✓	Ground
[33]	Scientometric	VOSviewer	✓	✓	✓	Aerial
[34]	Bibliometric	VOSviewer	✓	✓	✓	Aerial
Proposed Study	Scientometric and SLR	Both Tool	✓	✓	✓	All domain

virtual environments like the metaverse—an advancement that could inspire novel human-machine collaboration approaches in defense settings [41]. Table I shows the comparison of relevant articles about defense in all categories based on six parameters, such as approach, visualisation tool, domain categorisation, co-occurrence network analysis (CoNA), co-citation network analysis (CcNA), and research challenges.

A. Research Question

Numerous studies have explored the integration of emerging technologies such as AI, ML, IoT, DT, and 5G/6G networks in defense operations. However, there is a notable gap in the literature regarding a comprehensive scientometric analysis which systematically examines the research trends, collaboration networks, and technological advancements across ground, aerial, naval, and civil Defense. Addressing the gap is crucial to identify key contributors, research fronts, and emerging directions which shape future defense strategies. To bridge the gap, the following research questions have been formulated.

1) *RQ1:* What are the publication trends in defense from 2019 to 2024?
2) *RQ2:* What are the most frequently co-occurring keywords?
3) *RQ3:* Which documents are most co-cited in defense research?
4) *RQ4:* Which countries have the strongest research collaborations?
5) *RQ5:* What are the emerging trends and future directions in defense research.

III. RESEARCH METHODOLOGY

The current study employs a scientometric approach to investigate the evolutionary paradigms and future directions in defense research. Fig. 4 presents the methodological framework applied in this article, showcasing the systematic process used to visualize and analyse the literature across all defense domains.

A. Research Framework

The proposed research framework is structured into four key phases, each aimed at systematically addressing the objectives of the study.

1) *Introduction:* This article begins by introducing emerging technologies and the transformative impact on defense operations. Further, explores how advanced

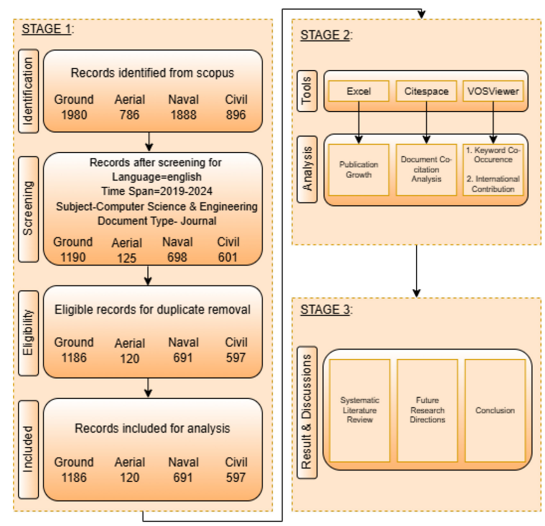

Fig. 4. Flow of the search methodology applied in the study.

technologies such as AI, ML, IoT, DT, and next-generation networks (5G/6G) are reshaping defense capabilities and enhancing the efficiency, accuracy, and responsiveness of defense services. The study examines the application of the technologies across all defense categories, including *Ground, Aerial, Naval, and Civil* defense. Additionally, this article introduces *a multi-layered defense architecture*, illustrating the end-to-end data flow process. This article also defines the research questions addressed in the study.

2) *Data Acquisition:* Equation (1) outlines the search query format used in the `(TITLE_ABS_KEY)` field of the Scopus database to retrieve datasets for each category. To extract data for a specific category, the relevant keywords are combined using OR operators, forming a comprehensive search string. The query is executed in the Scopus database to obtain matching records. To prevent overlap across categories, records belonging to other categories are excluded by applying the AND NOT operator along with the corresponding keyword sets

$$Q(i) = \left(K\_\mathrm{cat}_i \wedge \neg \bigcup_{\substack{j=1 \\ j \neq i}}^{n} K\_\mathrm{cat}_j \right). \tag{1}$$

where:

a) $Q(i) \rightarrow$ Represents the search query for the ith category;
b) $K\_\text{cat}_i \rightarrow$ The set of keywords corresponding to the current category; and
c) $n = 4 \rightarrow$ Total number of categories.

The categories are defined as

$$Q(1) = \text{“Ground”} \quad Q(2) = \text{“Aerial”}$$
$$Q(3) = \text{“Naval”} \quad Q(4) = \text{“Civil”}.$$

Fig. 4 presents the structured search strategy and article selection methodology, following the preferred reporting items for systematic reviews and meta-analyses (PRISMA) framework, which consists of four sequential phases. In the identification phase, initial queries were executed in the Scopus database using category-specific keyword sets in conjunction with emerging technology terms, such as IoT, DT, 4G, 5G, 6G, and AI. The Ground Defense query K_cat_1 included terms such as army and ground forces, aerial defense K_cat_2 included Air Force and “Air Combat,” Naval Defense K_cat_3 comprised terms like “Navy” and “Submarines,” while Civil Defense K_cat_4 incorporated keywords such as “Police” and “Homeland Security.” This phase yielded 1980 records for Ground, 786 for Aerial, 1888 for Naval, and 896 for Civil Defense. In the screening phase, filters were applied to refine the dataset based on the following criteria: Language = English, Publication Years = 2019–2024, Subject Area = Computer Science and Engineering, and Document Type = Journal. As a result, 1190 (Ground), 125 (Aerial), 698 (Naval), and 601 (Civil) records were retained. During the eligibility phase, duplicate entries were eliminated using network-based bibliometric visualisation tools to ensure data integrity and avoid overlap across domains. Finally, the included dataset used for analysis comprised 1186 records for Ground Defense, 120 for Aerial Defense, 691 for Naval Defense, and 597 for Civil Defense. Additionally, *20 records* were selected for the *SLR* using Rayyan. There are several tools available to support article selection in SLRs, including Colandr [42], JBI SUMARI [43], PICO Portal [44], Rayyan [45], SRDR+ [46], and SWIFT-Active Screener [47]. To support this article selection process for the SLR, Rayyan was employed as the primary screening tool. Articles were retrieved from Google Scholar with a focus on the application of AI and ML in defense domains. The inclusion criteria encompassed peer-reviewed journal articles and conference papers published between 2016 and 2025, written in English, and focused on the application of AI, ML, or emerging technologies within the domains of ground, aerial, naval, or civil defense. Using the criteria, Rayyan facilitated the collaborative screening process and helped refine the initial set of articles to 20 highly relevant references.

3) *Scientrometric Analysis:* Stage 2 involves a comprehensive analytical process, including *publication growth analysis*, *keyword CoNA*, *document co-citation analysis (DCA)*, and *country collaboration analysis*.
4) *Result and Discussion:* Stage 3 presents a *detailed SLR*, highlighting the *key findings* of the study. All *research questions* are addressed, followed by a summary of *future directions* and the *conclusion* of the study.

B. Databases and Tools Selection

The current paper presents a scientometric analysis of defense research within the field of *computer science and engineering*. Although various databases such as Scopus, Web of Science (WoS), and Dimensions are available for such analyses, this study utilises only the Scopus database. Scopus is the world’s largest abstract and citation database, indexing over 25 000 peer-reviewed journals, conference proceedings, and other scholarly sources. Furthermore, approximately 99.11% of the journals indexed in WoS are also covered by Scopus [48], ensuring broad and reliable coverage for scientometric analysis. IEEE Explore and ScienceDirect serve primarily as publisher-specific platforms for accessing journals and conference papers, rather than as comprehensive citation indexes. To support the analysis, two widely recognised visualisation tools are employed. *CiteSpace* is used for DCA and clustering of research themes, while *VOSViewer* is applied to map keyword associations and global research contributions. The tools facilitate the identification of major trends, collaborative networks, and thematic developments within the defense research landscape.

IV. Data Analysis and Interpretation

A. Research Dynamics in Defense

During 2019–2020, ground defense experienced a slight decline in publications *(dropping by about 6%),* which corresponded with a reduction in *citations* from 3276 to 2312. Meanwhile, aerial defense saw significant growth with publications increasing by *one-third* and citations more than doubling from *158 to 396*. Naval defense grew substantially by *30%,* reaching *2156 citations*, indicating a shift in research priorities. Civil defense maintained steady growth at *25%*, generating a considerable *1831 citations*. The *2021–2022* period showed even stronger trends. Ground defense rebounded strongly with nearly *38% publication* growth and *3208 citations*. Aerial defense recorded the most remarkable growth among all sectors at *87.5%*, though *citation* impact remained relatively modest at *406*. Naval defense continued steady growth at *16%*, achieving the *highest citation count (CC) of 2185*. Civil defense maintained consistent expansion with publication output increasing by *nearly 27%*. Ground defense continued expanding with an impressive *30%* growth in *2023–2024*, but *citations* dropped significantly from *2064 to 943*. Recent research has not had time to make an impact yet, or publication quality might have decreased. Aerial defense maintained strong momentum with *50% growth* in *publications* during *2023–2024*. However, *citations fell* dramatically from *282 to just 80*, raising questions about whether current research directions address relevant problems in the field.

TABLE II
PUBLICATION GROWTH AND CITATION ANALYSIS ACROSS ALL DEFENSE SECTORS

Year \ Category	Ground Defense			Aerial Defense			Naval Defense			Civil Defense		
	TP	**Growth (%)**	**Citation**	**TP**	**Growth (%)**	**Citation**	**TP**	**Growth (%)**	**Citation**	**TP**	**Growth (%)**	**Citation**
2019	118	-	3276	6	-	158	43	-	1201	55	-	1151
2020	111	-5.93%	2312	8	+33.33%	396	56	+30.23%	2156	69	+25.45%	1831
2021	153	+37.84%	3208	15	+87.50%	406	65	+16.07%	1617	75	+8.70%	2158
2022	203	+32.68%	2688	26	+73.33%	415	118	+81.54%	2185	95	+26.67%	1139
2023	261	+28.57%	2064	26	0.00%	282	169	+43.22%	2050	110	+15.79%	746
2024	340	+30.27%	943	39	+50.00%	80	240	+42.01%	696	103	-6.36%	210

TABLE III
MATHEMATICAL EQUATIONS FOR KEYWORD CO-OCCURRENCE, TLS, DEGREE, AND CENTRALITY CALCULATION

Equation No.	Equation	Equation Type	Significance
(1)	$C(i,j) = \sum_{D_k \in \mathcal{D}} \delta(K_i, D_k) \cdot \delta(K_j, D_k)$	Keyword Citation Count	Measures co-occurrence between keywords K_i and K_j across all documents in dataset $\mathcal{D}$. The function $\delta(K_i, D_k)$ returns 1 if K_i appears in D_k, otherwise 0.
(2)	$TLS(K_i) = \sum_{\substack{j=1 \\ j \neq i}}^{N} C(i,j)$	Total Link Strength	Computes the TLS of keyword K_i by summing all the co-occurrences with other keywords, indicating the influence in the network.
(3)	$D_i = \sum_{j \in N} A_{ij}$	Degree	Represents the number of direct connections (links) a reference has in the citation network. A higher degree indicates greater connectivity and influence.
(4)	$C_i = \sum_{j \neq i \neq k} \frac{\sigma_{jk}(i)}{\sigma_{jk}}$	Centrality	Measures the bridging role of a reference in connecting different research themes. Higher centrality suggests a key position in shaping knowledge flow.

Naval defense showed robust expansion at *42% in 2023–2024*, continuing the upward trend in research output. Despite growth, *citations dropped* substantially from *2050 to 696*, possibly indicating a saturation point where additional research is not generating new insights. Civil defense was the only sector to contract in *2023–2024*, with *publications declining by about 6%*. A sharp *drop* in *citations* to just *210* suggests the research area is losing relevance or focus in the current landscape. The analysis of defense sector research, displayed in Table II, reveals significant patterns in publication growth and citation impact across all defense categories.

B. Keyword Association Analysis

Keyword analysis is crucial in scientometric research for identifying trends, mapping knowledge domains, and predicting future research directions. By examining frequency, co-occurrence, and interrelationships, researchers gain insights into the evolution of a field and detect emerging research areas. Clustering related keywords highlights distinct subfields, while co-occurrence networks visually depict topic connections. VOSviewer offers three visualization layouts for keyword analysis. The *network visualization* illustrates keyword co-occurrence relationships and thematic clusters, helping identify how topics are interlinked within the research field. The *overlay visualization* highlights the temporal evolution of research topics, showcasing how themes have emerged and shifted over time. Lastly, the *density visualization* uses a heatmap effect to represent the concentration and prominence of keywords, indicating areas with significant research activity and impact. The network visualisation layout is used for keyword analysis, incorporating both *author keywords and index keywords* to ensure comprehensive domain coverage. The *full counting method* is applied instead of fractional counting, meaning each co-occurrence is counted in full rather than being divided, providing a more detailed representation of keyword relationships. A threshold value determines the selection of keywords from the *total set* N_k, based on the *minimum occurrence condition* M_k. *selected keywords* S_k appear as nodes, while edges (links) connect co-occurring keywords; thicker links indicate stronger relationships. Clusters, represented by different colors, indicate distinct research themes, with keywords in the same cluster sharing the same color. Links between clusters highlight interdisciplinary connections. Node size is determined by keyword occurrence and weight, where larger circles indicate higher weights. The distance between nodes signifies relationship strength, with shorter distances representing stronger associations. VOSViewer considers two essential metrics for analysing keyword co-occurrence networks: *CC and total link strength (TLS)*. CC denotes the frequency of a keyword's occurrence within the dataset, while TLS represents the total strength of associations with other terms, indicating the extent of co-occurrence with various keywords. A higher TLS suggests a keyword plays a central role in connecting various research topics. Table III presents the key parameters and the corresponding equations used in constructing the keyword co-occurrence network, highlighting the significance of the analysis.

1) Ground Defense: In the analysis of ground defense research, Out of *12 183 total keywords* (N_k) examined, *1314 keywords* (S_k) satisfied the *minimum occurrence* (M_k) *threshold of 3*. Fig. 9(a) shows the keyword co-occurrence network for Ground Defense, highlighting key themes like ML, network security, fault detection, and energy systems, reflecting AI-driven advancements in operational efficiency and threat monitoring. The network consists of *8 distinct* research clusters, *34 351 interconnected links*, and *52 298 TLS*. Fig. 5 illustrates the distribution of keywords across multiple thematic clusters within Ground Defense research. The figure highlights the diversity and density of research focus areas, reflecting how various technological and operational themes are grouped into distinct clusters based on keyword co-occurrence. The distribution supports the identification of

TABLE IV
COMPARISON OF NODE DETAILS IN GROUND, AERIAL, NAVAL, AND CIVIL DEFENSE DOMAINS

Ground Defense					Aerial Defense					Naval Defense					Civil Defense				
Node Name	CC	Degree	Centrality	TLS	Node Name	CC	Degree	Centrality	TLS	Node Name	CC	Degree	Centrality	TLS	Node Name	CC	Degree	Centrality	TLS
Machine Learning	280	312	0.29	2517	Reinforcement Learning	54	157	0.37	930	Deep Learning	293	335	0.19	3578	Machine Learning	131	192	0.27	792
Deep Learning	253	259	0.23	2226	Deep Learning	52	144	0.29	927	Autonomous Underwater Vehicles	261	304	0.15	3367	Deep Learning	126	191	0.26	895
Tanks (Containers)	153	257	0.15	1634	Air Combat	48	117	0.23	696	Machine Learning	187	307	0.17	2290	Crime	85	150	0.17	711
Learning Systems	140	242	0.13	1531	Reinforcement Learning	35	103	0.12	597	Autonomous Vehicles	126	247	0.07	1802	Artificial Intelligence	72	141	0.13	416
Artificial Intelligence	102	191	0.08	876	Decision Making	30	103	0.11	540	Learning Systems	106	200	0.08	1488	Law Enforcement	60	140	0.12	455

dominant and emerging topics in the field. Additionally, Table IV highlights the top five keywords, presenting the *CC, TLS, degree, and centrality metrics.*

1) *Cluster 1* includes one of the top five keywords, *Tank (Containers),* which has *153 occurrences* and a *TLS of 1385*. A high occurrence like 153 suggests significant relevance, while a TLS of 1385 indicates strong connectivity with other keywords, reinforcing the importance in the network. Other keywords, such as *Chemical Oxygen Demand*, *Temperature*, *Alkalinity*, and *Anaerobic Digestion* highlight environmental monitoring and resource management in defense. Meanwhile, terms like *Feeding*, *Antioxidants*, and *Tissues* suggest nutrition, health, and battlefield medicine. The keywords cluster together due to the shared role in ground defense logistics, sustainability, and soldier welfare.
2) *Cluster 2* includes the 5th top keyword, *AI,* which has *102 occurrences* and a *TLS of 1046*. While 102 occurrences indicate a strong research focus, a TLS of 1046 suggests significant interconnections with other terms, highlighting AI's central role in defense-related technologies. Keywords like *5G*, *5G Mobile Communication*, and *Antennas* emphasise next-generation connectivity, while *network security*, *botnets*, *cyberattacks*, and *blockchain* focus on cyber defense and secure communication. Additionally, *radio and circuit oscillations* relate to signal processing and electronic warfare. The keywords cluster together due to the collective role in enhancing defense communication, cybersecurity, and AI-driven operations.
3) *Cluster 3* includes the 2nd top keyword, *deep Learning,* with *253 keywory co-occurrences* and a *TLS of 2168*. A high occurrence count signifies strong research attention, while the high TLS value indicates extensive interconnections with other keywords, reinforcing deep learning's critical role in defense applications. Related keywords like *task analysis*, *feature analysis*, and *multiview stereo (MVS)* emphasise data processing and computer vision, while *fault analysis*, *fault detection*, and *fault diagnosis* highlight predictive maintenance and anomaly detection in defense systems. Additionally, *convolutional neural networks* plays a key role in image recognition and situational awareness, bringing the terms together under a common defense related research theme focused on AI-driven fault detection, predictive maintenance, and computer vision applications in defense.
4) *Cluster 4* focuses on *DTT* and DT applications in energy management and water systems. Keywords such as *digital storage*, *thermal storage*, *heating*, and *energy Consumption* highlight the role of energy optimisation and efficiency in defense operations. Additionally, terms like *energy loads*, *thermal energy*, and *waste heat* emphasise the importance of resource utilisation and sustainability. The presence of *water tanks*, *water distribution systems*, and *water heaters* indicates a focus on infrastructure resilience and operational readiness in defense settings. The cluster aligns with a research theme centred on DT integration for energy-efficient and sustainable defense infrastructure.
5) *Cluster 5* focuses on the human-centric aspect of defense research, integrating biological and technological components. Keywords like *Human*, *Male*, *Female*, and *Adult* indicate studies related to human factors, physiology, and performance assessment. The presence of *Human Experiment* and Major Clinical Study suggests research on cognitive, physical, and psychological evaluations in defense settings. Additionally, *Signal Processing* highlights the role of biomedical and neurophysiological data analysis, potentially for monitoring soldier health, brain activity, or stress levels. The cluster aligns with a research theme centered on human performance, physiological assessment, and signal processing in defense applications.

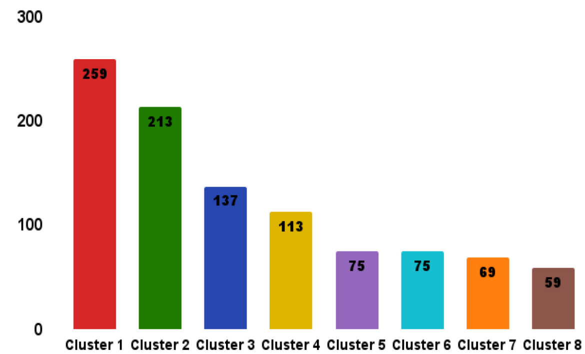

Fig. 5. Keyword distribution across clusters in ground defense.

2) Aerial Defense: In the analysis of aerial defense research, out of 1276 total keywords (N_k) examined, 54 keywords (S_k) satisfied the minimum occurrence (M_k) threshold of 3. Fig. 9(b) presents the keyword co-occurrence network for Aerial Defense, emphasising key research areas such as deep learning, reinforcement learning, UAVs, air combat, and autonomous decision-making, demonstrating a strong focus on AI-driven aerial operations and tactical automation. The network consists of 8 distinct research clusters, 1767 interconnected links, and 3834 TLSs. Fig. 6 visualises the keyword clusters in Aerial Defense research, highlighting the variety of technological themes and how keywords interrelate based on co-occurrence patterns. Table IV presents the top 5 keywords to highlight the most significant terms, accompanied

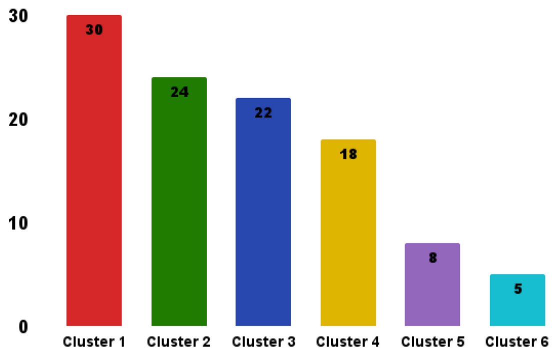

Fig. 6. Keyword distribution across clusters in aerial defense.

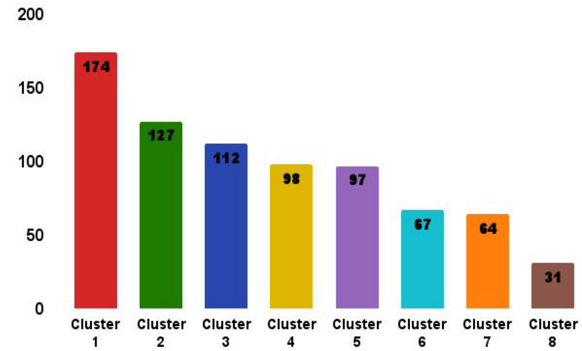

Fig. 7. Keyword distribution across clusters in naval defense.

by the respective CC, TLS, degree, and centrality metrics. The clusters are closely linked with each other.

1) *Cluster 1:* The first cluster in the aerial defense keyword network covers aviation operations, safety, AI applications, and environmental modelling. The cluster includes the top five keywords of the entire network. *reinforcement learning (54 citations)*, *deep learning (52)*, *air combat (48)*, *decision-making (30)*, and *UAV (26)*. Other key terms such as *air navigation*, *air traffic control*, *aircraft control*, *flight dynamics*, *flight simulations*, *aircraft accidents*, and *job analysis* contribute to optimising flight operations, safety, and training. Reinforcement learning and deep learning enhance autonomous decision-making and situational awareness, while air combat and UAVs play a critical role in modern aerial defense strategies.
2) *Cluster 2:* The second cluster focuses on air transportation, AI-driven decision making, predictive analytics, and security assessment. *Forecasting* and *optimization* enhance flight planning, while *anomaly detection*, *attention mechanisms*, and *decision support systems* improve risk management. *DTs* enable real-time simulation, and *ML*, *random forests*, and *time series* analysis support predictive modeling. *Threat assessment* strengthens security and defense strategies.
3) *Cluster 3:* The third cluster focuses on autonomous aerial systems and decision-making. *UAVs*, *antennas*, and *aerial vehicles* enable communication and navigation, while *autonomous agents* and *swarm intelligence* help coordinate and adapt to changing conditions. Complex networks and decision modelling improve planning and problem solving, and *RNNs* enhance predictive capabilities. Together, the technologies advance UAV autonomy for defense, surveillance, and logistics.
4) *Cluster 4:* The fourth cluster focuses on air combat and autonomous decision making. *Aerial vehicles* and *maneuver decision making* are essential for tactical operations, while *deep reinforcement learning* enhances adaptive strategies and real time responses. *Expert knowledge* supports informed decision making, improving mission effectiveness. The elements collectively support the advancement of intelligent, autonomous combat systems designed for aerial defense and military applications.
5) *Cluster 5:* The fifth cluster is centered around decision making and strategy in aerial defense. *Game theory* and *Nash equilibrium* help model competitive and cooperative tactics, while *actor–critic* and *learning algorithms* improve how systems adapt and make decisions. *Computation theory* ensures efficient problem-solving, and close-range interactions focus on quick tactical responses. The keywords come together to support the development of intelligent and strategic aerial systems.

3) Naval Defense: In the analysis of naval defense research, out of 5892 total keywords (N_k) examined, 770 keywords (S_k) satisfied the minimum occurrence (M_k) threshold of 3. Fig. 9(c) visualises the keyword co-occurrence network for Naval Defense, showcasing key research areas including deep learning, autonomous systems, maritime surveillance, object detection, and fault diagnosis, reflecting a concentrated focus on AI-driven technologies for navigation, monitoring, and system reliability in naval operations. The network comprises 8 distinct clusters, 28 385 interconnected links, and a TLS of 52 997. Fig. 7 illustrates the distribution of keywords across multiple thematic clusters within Naval Defense research. The figure highlights the diversity and density of research focus areas, reflecting how various technological and operational themes are grouped into distinct clusters based on keyword co-occurrence, while Table IV highlights the top 5 keywords, along with their CC, TLS, degree, and centrality metrics. The clusters exhibit strong interconnections, reflecting a cohesive research network.

1) *Cluster 1* is centred around AI-enabled maritime surveillance and object detection, with *Deep Learning* as the top keyword, having *293 occurrences* and a *TLS of 3578*. The high occurrence indicates substantial research interest, while the strong TLS suggests deep interconnections within the domain. The keywords within the cluster: *Object Detection*, *Feature Extraction*, *Maritime Surveillance*, *Detection Methods*, *Image Coding*, *Image Fusion*, I*image Processing*, *Object Recognition*, and *Ship Detection* are closely related as all contribute to enhancing situational awareness in naval operations. Deep Learning, combined with image

processing techniques, facilitates automated target detection, vessel tracking, and maritime threat identification. The cluster primarily focuses on leveraging AI and advanced computer vision to enhance real-time monitoring, anomaly detection, and naval security, representing a critical research area in defense technology.

2) *Cluster 2* focuses on AI-regulated fault detection and predictive maintenance, with *ML (3rd top keyword, 187 occurrences, 2290 TLS*) and *Learning Systems (5th top keyword, 106 occurrences, 1488 TLS)* playing a central role. The presence of keywords like *Fault Detection*, *Fault Diagnosis*, *Fault Tolerance*, *Feature Selection*, and *Neural Networks* highlights the emphasis on identifying, predicting, and mitigating system failures in naval operations. *ML* and *Learning Systems* enable automated anomaly detection, adaptive fault response, and decision-making for system reliability. The prime focus of the cluster is to enhance naval defense resilience through intelligent fault analysis and Intelligent technology-based predictive maintenance strategies, ensuring optimal system performance and operational safety.
3) *Cluster 3* is centered on autonomous navigation and control in underwater and surface vehicles, with *autonomous underwater vehicles (AUVs) (2nd top keyword, 261 occurrences, 3367 TLS*) and *Autonomous Vehicles (4th top keyword, 126 occurrences, 1802 TLS)* leading the research focus. The high occurrence and TLS values reflect strong academic and practical interest, emphasising the crucial role of autonomy in naval operations. The presence of keywords such as *Deep Reinforcement Learning*, *Path Finding*, *Path Following Control*, *Motion Control*, *Simulation Training*, and *Path Tracking* highlights advancements in AI-controlled navigation, real-time decision making, and motion optimisation. The technologies enable autonomous systems to efficiently traverse complex maritime environments, adapt to dynamic conditions, and enhance mission effectiveness. The prime focus of the cluster is to advance intelligent control mechanisms for autonomous naval vehicles, ensuring precise movement, obstacle avoidance, and strategic mission execution.
4) *Cluster 4* focuses on Underwater Sensor Networks and Wireless Communication Technologies essential for maritime surveillance, environmental monitoring, and submarine operations. Keywords such as *Underwater Sensor Network*, *Wireless Communication*, *Wireless Sensor Networks*, and *Sensor Nodes* highlight research on real-time data transmission, networking efficiency, and remote sensing in underwater environments. The inclusion of terms like *Underwater Object*, *Underwater Environment*, and *Submarine Cables* suggests a strong emphasis on detecting, tracking, and communicating with submerged assets, including vessels, marine life, and infrastructure. The prime focus of the cluster is to enhance underwater connectivity, improve sensor-based situational awareness, and develop resilient communication networks for naval operations.

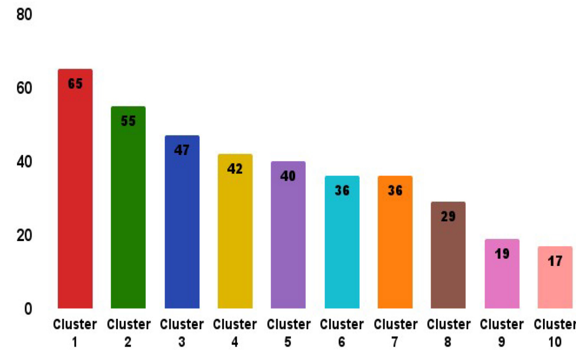

Fig. 8. Keyword distribution across clusters in civil defense.

5) *Cluster 5* focuses on the impact of climate change and advanced geophysical analysis on naval operations. The presence of keywords such as *Climate Change*, *Climate Methods*, and *Sea Surface Temperature* highlights research on environmental monitoring and its effects on maritime activities, including naval navigation, submarine operations, and oceanic warfare strategies. Keywords such as *Submarines Geophysics*, *Time Series*, and *Satellite Data* indicate a focus on analysing oceanic and atmospheric changes using remote sensing and data-driven modelling techniques. The inclusion of *Support Vector Regression* and *Convolution* suggests the application of ML models for predictive analytics in climate studies and naval simulations. The prime focus of the cluster is to enhance naval defense strategies by integrating climate data analysis, geophysical modelling, and AI-driven forecasting for improved decision-making in maritime security.

4) Civil Defense: In the analysis of civil defense research, out of 4123 total keywords (N_k) examined, 386 keywords (S_k) satisfied the minimum occurrence (M_k) threshold of 3. Fig. 9(d) displays the keyword co-occurrence network for Civil Defense, emphasising core research areas such as ML, deep learning, crime prediction, accident detection, and smart surveillance, highlighting the growing application of AI technologies in enhancing public safety, urban monitoring, and law enforcement systems. The network comprises 10 distinct clusters, 8501 interconnected links, and a TLS of 12857. Fig. 8 illustrates the distribution of keywords across thematic clusters in Civil Defense research, reflecting the diversity of focus areas. Higher keyword counts in clusters indicate more prominent research themes, while smaller clusters suggest emerging or niche topics. Table IV highlights the top 5 keywords, along with their CC, TLS, degree, and centrality metrics. The clusters exhibit strong interconnections, reflecting a cohesive research network.

1) *Cluster 1* focuses on advanced communication and smart infrastructure, incorporating key technologies such as *5G and IoT*. The presence of *5G mobile communication* and *vehicle-to-vehicle communication* highlights the role of high-speed, low-latency networks in enabling real-time data exchange. Keywords like *accident detection*

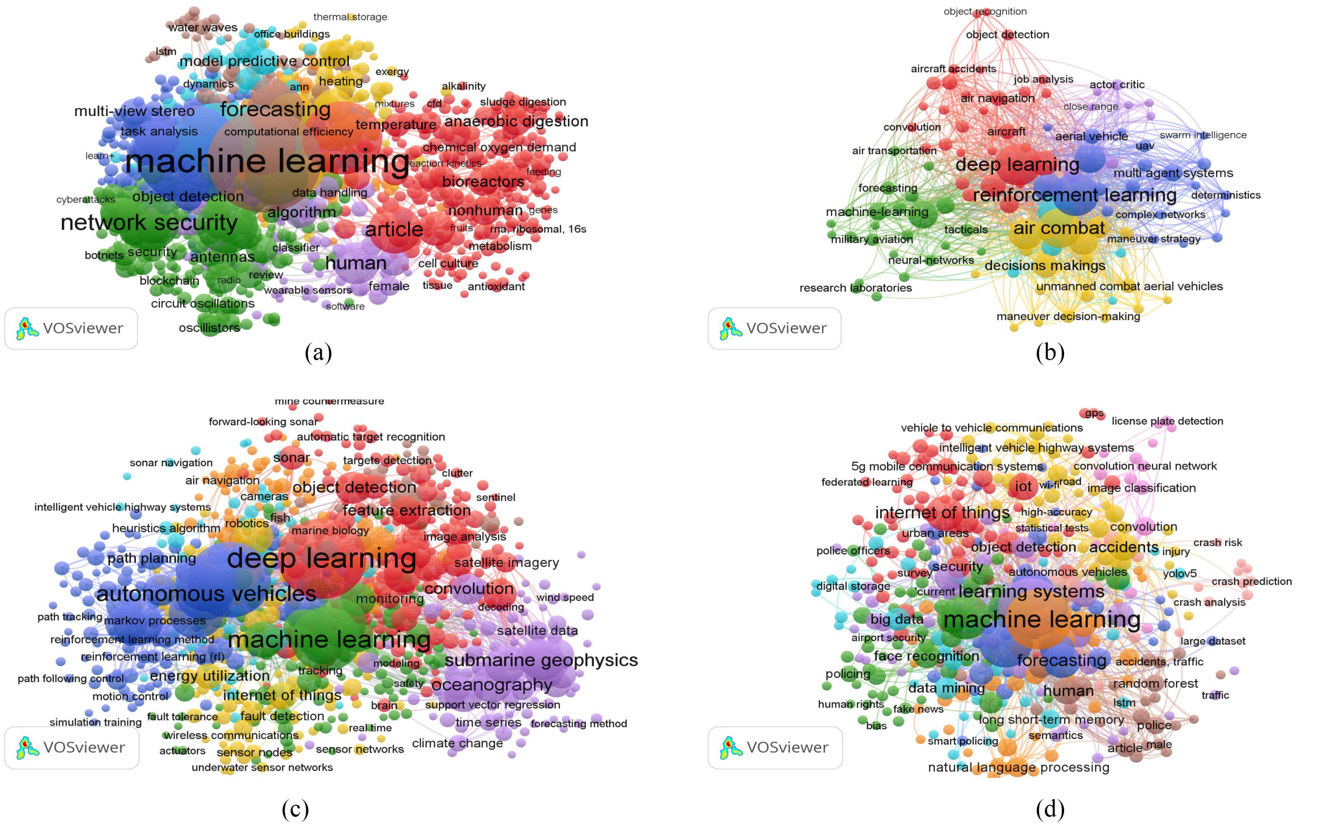

Fig. 9. Keyword co-occurrence analysis across different defense sectors. (a) Keyword co-occurrence network in ground defense. (b) Keyword co-occurrence network in aerial defense. (c) Keyword co-occurrence network in naval defense. (d) Keyword co-occurrence network in civil defense.

emphasise applications in traffic safety, while *federated learning* suggests privacy-preserving AI models for decentralised data processing. The cluster primarily revolves around leveraging 5G and IoT for intelligent transportation and emergency response systems in civil defense.

2) *Cluster 2* is centered around AI and *Law Enforcement*, featuring *AI (4th top keyword)* with *72 occurrences and 416 TLS*, and *Law Enforcement (5th top keyword) with 60 occurrences and 455 TLS*. The presence of keywords like *policing*, *fake news*, *human rights*, *bias*, and *face recognition* indicates a focus on the role of AI in modern law enforcement. The cluster highlights ML-based policing, digital forensics, surveillance, and ethical considerations in law enforcement applications.
3) *Cluster 3* includes the 3rd top keyword, *Crime,* with *85 occurrences and 711 TLS*. The cluster is centred around textit forecasting, *crime prediction*, *crime detection*, *criminal networks*, and *data analysis*, highlighting the role of data-driven techniques in crime prevention. The focus is on predictive analytics, pattern recognition, and network analysis to enhance law enforcement strategies and improve public safety.
4) *Cluster 4* focuses on *accident prevention* and *road safety*, featuring keywords such as *accidents*, *anomaly detection*, *road*, *road safety*, *road detection*, and *roads and streets*. The cluster highlights research on traffic management, hazard detection, and safety enhancements through advanced monitoring and predictive techniques, aiming to reduce accidents and improve urban mobility.
5) *Cluster 5* includes the *2nd top keyword, Deep Learning, with 126 occurrences and 895 TLS*. A high occurrence count reflects significant research attention, while the strong TLS indicates deep interconnections with other terms. Related keywords such as *artificial neural network*, *convolutional neural network*, *deep neural network*, *image processing*, and *image enhancement* emphasise AI-driven techniques for image analysis, pattern recognition, and automated decision-making. The prime focus of the cluster is leveraging deep learning for advanced surveillance, anomaly detection, and security applications in civil defense.

C. Key Thematic Areas in Defense

A focused analysis across ground, aerial, and naval defense domains identifies several dominant thematic areas shaping military innovation. AI, ML, and DL are central to modern defense systems, enabling autonomous threat detection, fault diagnosis, and predictive maintenance. Autonomous platforms such as autonomous aerial, underwater, and ground vehicles are being developed for intelligent navigation, reconnaissance, and mission execution. Cybersecurity and resilient communication infrastructures, including blockchain, 5G, and underwater wireless networks, are critical for secure data

TABLE V
KEY PARAMETERS IN DCA

Parameter	Meaning	Significance	Mathematical Equation
Cluster Size	Number of citing references in the cluster	Larger size → a more established and influential research area	$CS = \sum_{i=1}^{n} C_i$
Mean Year	Average publication year of papers in the cluster	Higher value → indicates a recent and emerging research focus	$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$
LLR Labels	Keywords that define the cluster based on Log-Likelihood Ratio (LLR) method	Helps identify dominant research themes within the cluster	$LLR = 2\sum_{i=1}^{n}\left(O_i \ln \frac{O_i}{E_i}\right)$
Citation Nodes	Number of unique cited papers within the cluster	More nodes → stronger knowledge foundation	$CN = \|N\|$, where N is the set of cited papers
Citation Counts	Total number of times cluster-related papers are cited	Higher citations → greater impact and research visibility	$CC = \sum_{i=1}^{n} C_i$
Degree	Number of connections (links) per cited paper	Higher degree → greater influence and connectivity in the network	$D_i = \sum_{j \in N} A_{ij}$, where A_{ij} is adjacency matrix entry
Centrality	Betweenness centrality of a cited paper, measuring the bridging role	High centrality → key research paper linking different topics	$C_i = \sum_{j \neq i \neq k} \frac{\sigma_{jk}(i)}{\sigma_{jk}}$
Silhouette Value (S)	Measures how well a paper fits within the cluster	High S (close to +1) → well-defined cluster with strong cohesion; S near 0 → overlapping clusters; Negative S → poor clustering	$S(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}$

TABLE VI
CLUSTER ANALYSIS RESULTS FROM DOCUMENT CO-CITATION

Ground Defense

ClusterID	Size	Silhouette	Label (LLR)	Avg. Year
0	51	0.91	multi-view stereo (48.79, 1.0E-4)	2013
1	33	0.934	using machine (58.39, 1.0E-4)	2013
2	28	0.909	multi-view stereo (51.12, 1.0E-4)	2018
3	26	0.935	normal consistency (42.17, 1.0E-4)	2016
4	26	0.964	industrial processes (43.83, 1.0E-4)	2013
5	24	0.832	object detection (49.12, 1.0E-4)	2013
8	14	0.973	botnet defense system (55.56, 1.0E-4)	2017

Naval Defense

ClusterID	Size	Silhouette	Label (LLR)	Avg. Year
0	49	0.923	deep reinforcement learning (58.7, 1.0E-4)	2018
1	39	0.779	adaptive deep RL (38.83, 1.0E-4)	2021
2	34	0.916	ship detection (44.87, 1.0E-4)	2020
3	31	0.914	multi-tier edge computing (38.02, 1.0E-4)	2019
4	23	0.925	adversarial fusion network (46.11, 1.0E-4)	2019
5	18	0.993	vessel trajectory prediction (30.93, 1.0E-4)	2017
6	17	0.983	review (26.86, 1.0E-4)	2019
7	16	0.987	enhancing Doppler velocity (26.4, 1.0E-4)	2019
8	15	1.000	sea surface temperature (38.87, 1.0E-4)	2018
9	12	0.957	fault diagnosis (30.59, 1.0E-4)	2020
13	7	0.997	predictive controller (12.18, 0.001)	2016

Aerial Defense

ClusterID	Size	Silhouette	Label (LLR)	Avg. Year
0	52	0.896	multi-aircraft close-range air combat (8.89, 0.005)	2020
1	33	0.926	multi-agent hierarchical policy gradient (17.34, 1.0E-4)	2017
2	27	0.986	CNN target detection (12.53, 0.001)	2015
3	26	0.948	object detection (11.46, 0.001)	2017
4	25	0.948	multi-aircraft combat (16.98, 1.0E-4)	2019
5	22	0.992	wildfire surveillance (8.71, 0.005)	2016

Civil Defense

ClusterID	Size	Silhouette	Label (LLR)	Avg. Year
0	41	0.993	using machine learning (23.78, 1.0E-4)	2006
1	22	0.935	big data (25.83, 1.0E-4)	1990
3	17	0.869	systematic review (35.46, 1.0E-4)	2006
4	17	0.963	comparative analysis (23.14, 1.0E-4)	2012
6	16	0.968	deep learning architecture (13.12, 0.001)	1976
7	15	0.956	learning approach (19.35, 1.0E-4)	2012

transmission in network-centric warfare. DTT and simulations support real-time modelling, mission rehearsal, and system optimisation for operational readiness. Situational awareness and surveillance are enhanced through satellite imagery, sensor fusion, and AI-driven object tracking for superior tactical intelligence. Efforts in energy and thermal management aim to support sustainable operations in extreme defense environments. Human-centric defense systems, including soldier health monitoring and human-machine interfaces, strengthen decision support and combat effectiveness. Lastly, intelligent combat modelling and decision-making systems, leveraging reinforcement learning and expert systems, enable adaptive responses in complex battlefield scenarios.

D. Document Co-Citation Analysis

DCA, introduced by Small (1973), examines how frequently two articles are cited together in the reference lists of other publications, thereby establishing a co-citation relationship. DCA in CiteSpace is a scientometric technique used to identify influential papers and research trends in a specific domain. The structural quality of a co-citation cluster is evaluated based on *Modularity (Q)*, which measures the separation between clusters, and *Weighted Mean Silhouette*, which assesses the cohesion and consistency within clusters. The *Q score ranges from 0 to +1*, with higher values indicating well-structured clusters. The *Weighted Mean Silhouette score ranges from −1 to +1*, where *values close to +1 suggest well-defined clusters*, *0 indicates overlap*, and *−1 implies poor clustering*. There are several additional parameters which contribute to evaluating the network quality in DCA, including cluster size, mean year, log-likelihood ratio (LLR) labels, citation nodes, CCs, degree, and centrality. Table V presents the definitions, significance, and corresponding mathematical equations for each of the parameters, while Tables VI and VII offer comprehensive details on *cluster size, silhouette value (S), average year, LLR labels, citation nodes, CCs, degree, and centrality*.

1) Ground Defense: The network consists of *415 nodes and 1284 edges*, with a *density of 0.0149*. The selection criteria include *g-index (K = 40), LRF = − 1.0, L/N = − 1, LBY = − 1, and e = 1.0*. Seven clusters are formed, with the focus directed toward analysing the top three. Cluster labels are derived using the LLR algorithm, the default method in CiteSpace. The network quality is reflected by a *Q value of 0.7887*, indicating well-separated clusters, and a *Weighted Mean Silhouette score of 0.9192*, signifying high internal consistency and minimal overlap. The analysis spans the years *2019–2024*, divided into *one-year slices*. Fig. 10(a) illustrates major research clusters in Ground Defense, including MVS, object detection, and botnet defense, reflecting the field's interdisciplinary focus and technological advancement. **Cluster#0**

TABLE VII
COMPREHENSIVE ANALYSIS OF CITATION IN DEFENSE CATEGORIES

Ground Defense

Category	Cluster ID	Citation Node	Citation	Degree	Centrality
Ground	0	Goodfellow I, 2016	25	51	0.16
Ground	2	Kingma DP, 2014	19	44	0.12
Ground	0	Mnih V, 2015	8	34	0.10
Ground	0	Hochreiter S, 1997	14	31	0.07
Ground	3	Knapitsch A, 2017	12	31	0.04
Ground	0	Sutton RS, 2018	12	26	0.04
Ground	1	Pedregosa F, 2011	10	25	0.03
Ground	1	Hochreiter S, 1997	10	24	0.03
Ground	4	LeCun Y, 2015	9	21	0.03
Ground	5	Simonyan K, 2014	9	21	0.03

Aerial Defense

Category	Cluster ID	Citation Node	Citation	Degree	Centrality
Aerial	0	Vinyals O, 2019	9	43	0.35
Aerial	0	Yang Q, 2019	9	39	0.20
Aerial	0	Hu D, 2021	8	36	0.17
Aerial	1	Silver D, 2017	8	34	0.16
Aerial	0	Yang Q, 2020	8	33	0.11
Aerial	0	Sutton RS, 2018	6	33	0.06
Aerial	4	Sun Z, 2021	5	30	0.06
Aerial	0	Wang M, 2020	5	27	0.06
Aerial	5	Mnih V, 2015	4	26	0.05
Aerial	3	Schulman J, 2017	4	26	0.04

Naval Defense

Category	Cluster ID	Citation Node	Citation	Degree	Centrality
Naval	0	Sutton RS, 2018	16	37	0.27
Naval	2	Redmon J, 2018	15	36	0.21
Naval	0	Fang Y, 2022	10	25	0.20
Naval	0	Zhang Q, 2020	9	24	0.16
Naval	2	Bochkovskiy A, 2020	23	31	0.16
Naval	0	Cheng C, 2021	7	23	0.11
Naval	0	Sun Y, 2020	7	22	0.11
Naval	3	Lin C, 2020	7	21	0.10
Naval	0	Schulman J, 2017	7	19	0.10
Naval	8	Xiao C, 2019	7	19	0.09

Civil Defense

Category	Cluster ID	Citation Node	Citation	Degree	Centrality
Civil	4	Safat W, 2021	10	32	0.07
Civil	0	Redmon J, 2018	9	23	0.06
Civil	0	Simonyan K, 2014	8	20	0.05
Civil	0	He K, 2016	8	17	0.05
Civil	1	ONeil C, 2016	8	17	0.05
Civil	7	Breiman L, 2001	6	17	0.04
Civil	4	Stalidis P, 2021	5	17	0.04
Civil	0	Goodfellow I, 2016	5	17	0.04
Civil	6	Perry WL, 2013	5	17	0.04
Civil	0	Hochreiter S, 1997	5	17	0.03

is the largest cluster with *51 members*, an *S value of 0.91*, and a *mean publication year of 2013*.Cluster is associated with the LLR theme *MVS (LLR = 48.79, p = 1.0E-4)*. The key reference, Goodfellow et al. 2016 [49], addresses adversarial vulnerabilities in neural networks, introducing the fast gradient sign method (FGSM) for adversarial sample generation, highlighting the need for resilient neural network models in defense systems. **Cluster#1**, the second-largest, includes *33 members* with an *S value of 0.934* and aligns with the LLR theme *using machine (LLR = 58.39, p = 1.0E-4)*. The key study by Hu and Tan 2017 [50], introduced MalGAN, a GAN-based method that generates adversarial malware capable of evading closed box ML detection systems, shedding light on cybersecurity concerns within ground defense. **Cluster#2** consists of *28 members* with an *S value of 0.909* and corresponds to the LLR theme *weak texture (LLR = 42.17, p = 1.0E-4)*. Key references include Giang et al. 2021 [51], who proposed CDSFNet to improve 3-D reconstruction by dynamically selecting optimal patch scales based on image curvature, and Wang et al. 2021 [52], who developed PatchmatchNet—an efficient, memory-conscious MVS technique suitable for constrained environments. All clusters display statistically significant LLR values *all* $>$ 10 *and p-value* $<$ 0.05, confirming the thematic coherence and robustness of the clustering. Collectively, the clusters contribute significantly to ground defense applications by enabling fast, accurate, and resource-efficient 3-D reconstruction vital for terrain mapping, surveillance, and target detection.

2) Aerial Defense: The network consists of *308 nodes and 1476 edges*, with a *density of 0.0312*. The selection criteria include *g-index (K = 40), LRF = – 1.0, L/N = – 1, LBY = 5, and e = 1.0*. The analysis identifies the six distinct clusters, with the study focusing on the top three clusters. Cluster labels are extracted using the LLR algorithm, which is the default method in CiteSpace. LLR provides broad topic coverage and ensures unique labelling for each cluster. The overall network quality is indicated by *Q value of 0.7917* and *Weighted Mean Silhouette score of 0.9403*, signifying a well-structured and high-quality clustering result. The analysis spans the *2019–2024* period, with a slice length of *1 year* per time interval. Fig. 10(b) shows key research clusters in Aerial Defense, focusing on multiagent systems, target detection, and air combat. The figure reflects the integration of AI and deep learning in autonomous aerial operations. **Cluster#0** is the largest cluster consist of *52 members with S = 0.896*. The *mean year of the cluster is 2020*. The prime reference of Cluster#0 is Yang et al. 2019 [53], which explored autonomous maneuver decision making, reinforcement learning in UAV combat, and DQN-based tactical decisions. Similarly, another key reference Vinyals et al. 2019 [54], introduced AlphaStar, a multiagent reinforcement learning system for complex, real-time strategy optimisation in StarCraft II, demonstrating advanced multiagent coordination and decision making. Both studies align with the core themes labelled multiaircraft close-range air combat. The *LLR value of 8.89 (p = 0.005)* statistically validates the label, indicating its strong relevance to the cluster. Emphasis is placed on autonomous tactics, strategic coordination, and real-time maneuvering in dynamic aerial engagements. **Cluster#1** is second largest cluster has *33 members with S=0.926*. The major reference Yang et al. 2020 [55] autonomous evasived maneuver strategies for autonomous Combat Aerial Vehicles facing beyond visual range missile threats, utilising multiobjective optimisation and hierarchical decision making to enhance evasion effectiveness. Second key reference Shin et al. 2018 [56] explored autonomous aerial combat algorithms for within visual range engagements, incorporating energy maneuver theory, cooperative decision making, and real-time simulation validation. Both studies align with the LLR theme multiagent hierarchical policy gradient *(LLR: 17.34, p-value: 1.0E-4)*, where the high LLR value indicates a strong association with the cluster, and the low p-value confirms statistical significance. Advancements in multiagent coordination, hierarchical learning, and tactical decision-making in aerial combat are highlighted. **Cluster#2** ranks third with *27 members and an S value of 0.986*. The key references, Hu et al. 2021 [57] and Paoletti et al. 2018 [58], explored CNN-based approaches for remote sensing image classification. Hu D. investigates transferring pretrained

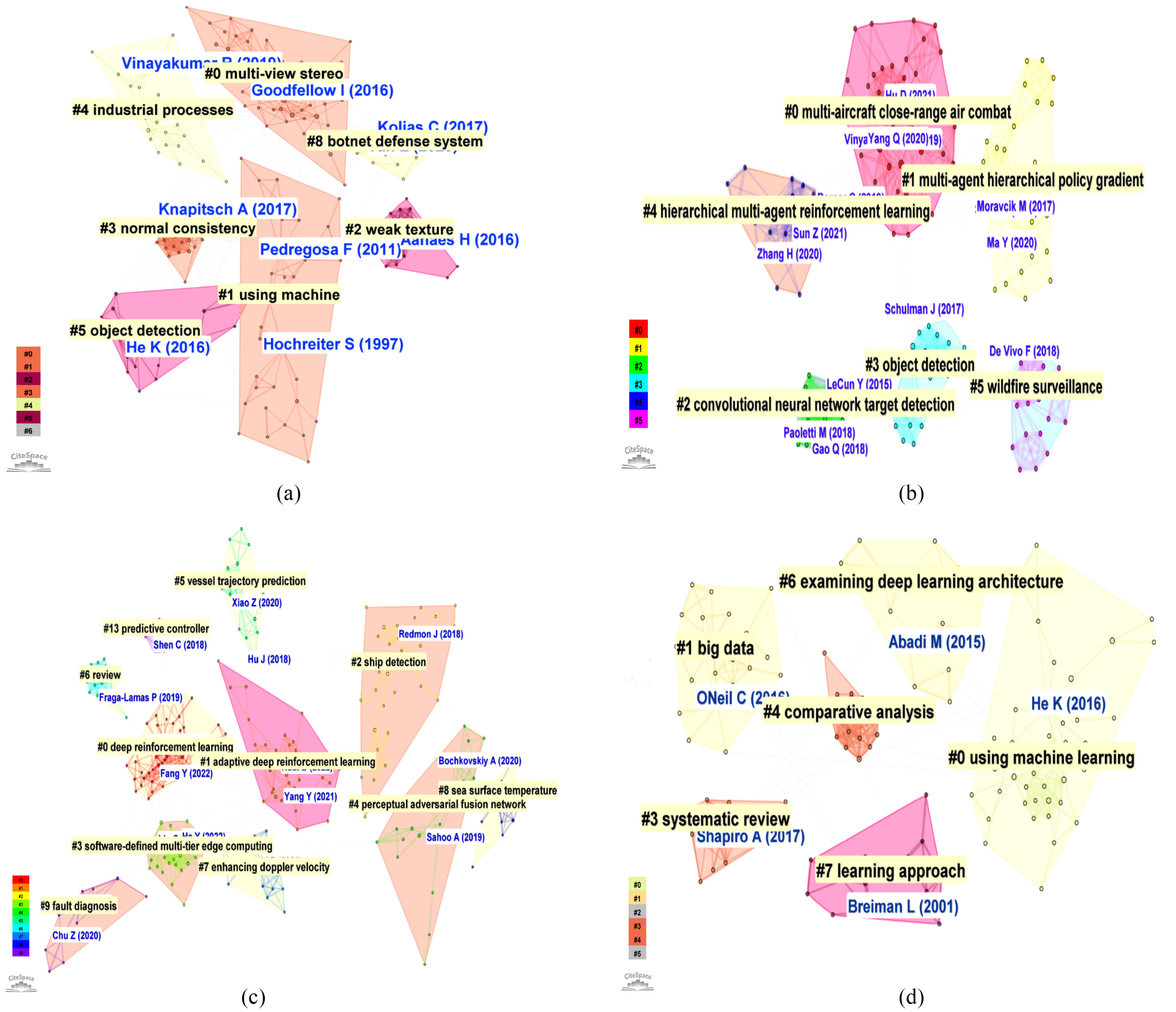

Fig. 10. DCA across different domains. (a) Cluster analysis in ground defense. (b) Cluster analysis in aerial defense. (c) Cluster analysis in naval defense. (d) Cluster analysis in civil defense.

CNN features for high-resolution remote sensing, proposing fully connected layer activations and multiscale convolutional feature encoding. Paoletti M. focuses on hyperspectral image classification using a 3-D CNN architecture, which integrates spectral spatial information and border mirroring for enhanced accuracy and efficiency. The studies align with the LLR theme convolutional neural network target detection *(LLR: 12.53, p-value: 0.001)*, reinforcing the role of CNNs in advancing remote sensing classification and target detection. All three clusters exhibit LLR and p-values which surpass the threshold, with *LLR* $>$ 10 *and p-value* $<$ 0.05, indicating strong statistical significance and highly cohesive clusters.

3) Naval Defense: The network comprises *401 nodes and 1194 edges*, with *a density of 0.0149*. The selection criteria include *g-index* $(K = 40)$, *LRF* $= -1.0$, *L/N* $= -1$, *LBY* $= 5$, *and e* $= 1.0$. The network is divided into eleven clusters, with the analysis focusing on the top three clusters. Cluster labels are generated using the LLR algorithm and ensure unique labelling for each cluster. The network quality is reflected by a *Q value of 0.7937* and *a Weighted Mean Silhouette score of 0.9203*, indicating a well-structured and high-quality clustering result. The study spans from *2019 to 2024*, with a *1 year* time slice for each interval. Fig. 10(c) highlights key research clusters in Naval Defense, emphasising themes like deep reinforcement learning, ship detection, and predictive control. The visualisation reflects focused research efforts on AI-driven autonomy, surveillance, and maritime system optimisation. **Cluster#0** is the largest cluster with *49 members* and *a silhouette value (S) of 0.923*, indicating high internal consistency. The *average year is 2017*, highlighting the recent relevance. The prime reference, Cui et al. 2017 [59], introduced a DRL-based adaptive control system for AUV, enhancing trajectory tracking under external disturbances and model uncertainties. Another key reference, Fujimoto et al. 2018 [60] presented a double *Q*-learning-based actor–critic framework, reducing overestimation bias and improving policy optimisation. The cluster aligns with the LLR theme deep reinforcement learning *(LLR: 58.7, p-value: 1.0E-4)*, indicating a strong thematic cohesion and high statistical significance. The LLR value of *58.7 reflects a highly cohesive cluster*, while the *p-value of 1.0E-4* confirms the association is statistically reliable and far below the standard *threshold of 0.05*. The combined research advances naval defense by enhancing AUV navigation, maritime surveillance, and underwater reconnaissance with DRL-driven precision and adaptability. **Cluster#1** is the second largest cluster with *39 members and S = 0.779*, indicating moderate internal consistency. The *average year is 2021*. The prime reference, Okereke et al. 2023 [61] presented an overview of ML techniques for AUV local path planning, addressing challenges such as safety, obstacle avoidance, energy efficiency, and navigation optimisation. The study explores supervised, unsupervised, and reinforcement learning approaches, analysing the effectiveness in real life and simulated scenarios. The cluster aligns with the LLR theme adaptive deep reinforcement learning *(LLR: 38.83, p-value: 1.0E-4)*, showing a strong and statistically significant association. The combined research advances naval defense by enhancing AUV performance with adaptive DRL techniques, enabling precise, efficient, and resilient underwater navigation. **Cluster#2** is the third largest cluster with *34 members* and a *silhouette value (S) of 0.916*, indicating high consistency. The *mean year is 2020*. The prime reference, Lang et al. 2022 [62] presented a multisource heterogeneous transfer learning method to enhance ship classification in SAR imagery by leveraging multiple source domains. The cluster aligns with the LLR theme ship detection *(LLR: 42.91, p-value: 1.0E-4)*, indicating a strong and statistically significant association. All three clusters exhibit *LLR* and *p-values* that surpass the threshold, with *LLR* $>$ 20 *and p-value* $<$ 0.05, indicating strong statistical significance and highly cohesive clusters. The cluster demonstrates relevance to naval defense through applications in maritime surveillance, ship classification, and border security operations.

4) Civil Defense: The network consists of *349 nodes and 886 edges*, with *a density of 0.0146*, representing a moderately connected structure. The selection criteria include *g-index (K = 40), LRF = −1.0, L/N = −1, LBY = −1, and e = 1.0*. The network is divided into six clusters, with the analysis focusing on the top three due to their higher significance and influence. Cluster labels are assigned using the LLR algorithm. The network quality is validated by a *Q value of 0.8546* and *a Weighted Mean Silhouette score of 0.9555*, indicating a well-structured, reliable, and high-quality clustering result. The *Q* value reflects the network's modularity, demonstrating clear separation between clusters, while the high silhouette score signifies strong internal cohesion and distinct boundaries, enhancing the reliability of the clustering. The analysis *spans the period from 2019 to 2024*, with a *1-year* time slice per interval. Fig. 10(d) highlights key research clusters in Civil Defense, including ML, big data, systematic reviews, and learning approaches, reflecting the application of data-driven methods in civil security and emergency management. **Cluster#0** is the largest cluster with *41 members and S=0.993*, indicating exceptionally high consistency. The *mean year of 2006* indicates foundational relevance within the domain. The prime reference, He et al. 2016 [63] introduced deep residual learning, enhancing the training of very deep networks by learning residual functions. It improves accuracy, achieving 3.57% error on ImageNet and a 28% boost on COCO detection. The cluster aligns with the LLR theme deep residual learning *(LLR: 45.62 > threshold 20, p-value: 1.0E-4 < threshold 0.05)*, indicating strong statistical significance. The research strengthens naval defense by enhancing image classification, target recognition, and maritime surveillance.

Cluster#1 is the second largest cluster with *22 members and S = 0.935*, indicating high internal consistency. The *average year is 2020*, highlighting the recent relevance. The prime reference, Brayne 2017 [64], examined the impact of big data analytics on police surveillance, revealing how it amplifies prior practices and transforms law enforcement. The study highlights predictive analytics, automated alert systems, and integrated data networks, enabling large-scale and proactive surveillance. The cluster aligns with the LLR theme big data surveillance *(LLR: 40.12 > threshold 20, p-value: 1.0E-4 < threshold 0.05)*, indicating strong statistical significance. The research enhances naval defense by advancing predictive threat detection, risk assessment, and large-scale maritime surveillance. **Cluster#3** is the third largest cluster with *17 members and a silhouette value (S) of 0.869*, indicating high consistency. The *mean year is 2006*. The key reference, Bhatti et al. 2021 [65] presented a real-time weapon detection system using CCTV footage and deep learning algorithms. Models such as YOLOv4 are employed, achieving the highest F1-score of 91% and a mean average precision of 91.73%, thereby outperforming previous methods. The cluster aligns with the LLR theme systematic review *(LLR: 39.72, p-value: 1.0E-4)*, exceeding the *LLR threshold of 20 and the p-value threshold of 0.05*, indicating strong statistical significance. The research enhances naval defense by improving real-time threat detection and surveillance efficiency through automated weapon identification.

5) Evaluation and Critical Analysis of Key Sources Within Research Clusters: As detailed in Table I, the related work section offers a comparative overview of foundational studies, highlighting the focus, methodologies, and defense applicability. Building on this, the DCA across Ground, Aerial, Naval, and Civil Defense domains further dissects the literature by identifying thematic clusters based on co-occurrence patterns. Each domain's DCA map includes highly cited documents grouped into distinct clusters, reflecting concentrated research trends. Building on this, the DCA across ground, aerial, naval, and civil Defense domains further dissects the literature by identifying thematic clusters based on co-occurrence patterns. Each domain's DCA map includes highly cited documents grouped into distinct clusters, reflecting concentrated research trends. From the broader analysis, critical clusters have been identified and evaluated based on their relevance to defense technologies and strategic implementation. In the ground defense domain, a primary cluster focuses on AI-driven surveillance and situational awareness, featuring sources that advance real-time target recognition, multisensor fusion, and automated battlefield monitoring. The works contribute to

TABLE VIII
COMPARISON OF CITATION DETAILS ACROSS GROUND, AERIAL, NAVAL, AND CIVIL DEFENSE DOMAINS

Ground Defense				Aerial Defense				Naval Defense				Civil Defense			
Country	**Citation**	**TLS**	**Centrality**	**Country**	**Citation**	**TLS**	**Centrality**	**Country**	**Citation**	**TLS**	**Centrality**	**Country**	**Citation**	**TLS**	**Centrality**
China	3713	30	0.30	China	5478	104	0.54	China	5478	104	0.54	India	1107	29	0.15
United States	2524	12	0.25	United States	1228	32	0.20	United States	1228	32	0.20	United States	1285	27	0.20
India	1572	12	0.16	India	836	22	0.15	India	836	22	0.15	China	1034	21	0.25
United Kingdom	1494	6	0.26	Australia	403	12	0.10	Australia	403	12	0.10	Pakistan	448	18	0.17
Saudi Arabia	822	8	0.12	United Kingdom	704	38	0.18	United Kingdom	704	38	0.18	Canada	263	20	0.09

enhancing operational responsiveness and threat detection in combat scenarios. Within aerial defense, a major cluster centres on autonomous aerial systems, particularly UAVs, emphasising advancements in deep reinforcement learning, real-time tracking, and adaptive mission planning crucial for intelligence, surveillance, and reconnaissance (ISR) missions. The naval defense domain highlights clusters related to maritime object detection and autonomous navigation, with foundational works addressing deep-learning-based ship recognition, underwater motion control, and mission planning for AUVs. The clusters directly support strategic maritime operations, including surveillance, reconnaissance, and navigation in complex aquatic environments. In civil defense, critical contributions emerge from clusters focusing on AI-based law enforcement, crime prediction, and smart surveillance. The studies explore data-driven methods for anomaly detection, urban safety monitoring, and ethical considerations in public security technologies.

E. International Contribution

Advanced technologies, including AI, ML, IoT, and DT, have dramatically transformed national defense systems. The innovations provide better situational awareness, more effective decision-making, increased operational efficiency, faster communication, and cost savings. Integrating 5G/6G networks further elevates the capabilities by enabling rapid data transfer from devices like UAVs, supporting real-time analysis in critical situations. Recognising the advantages, nations are investing heavily to incorporate cutting-edge technologies into the defense strategies. The VOSViewer visualisation tool was employed to analyze the network of productive countries from *2019 to 2024, using a 1-year time* slice for detailed examination. The overlay visualization illustrates the temporal evolution of country collaborations by using *color-coded networks based on publication years*. A threshold value is applied to select countries from the *total set* T_C, based on two criteria: 1) *the minimum number of documents per country* M_D and 2) the *minimum number of citations per country* M_C. The *selected countries* S_C are represented as nodes in the visualisation. The links between country nodes signify collaborative relationships or co-authorship connections. Thicker or stronger links indicate a higher frequency of collaboration, whereas thinner or weaker links reflect fewer joint publications between the countries. The node size in the collaboration network signifies the influence and prominence of a country. Larger nodes represent countries with higher publication output, greater citation impact, and stronger centrality, indicating the key role in the research network. Centrality thresholds are applied to evaluate the influence of each country: *Centrality*$\geq$ 0.20 *indicates high influence* with major collaboration hubs, $0.10 \leq$ *Centrality* < 0.20 *represents moderate influence* with decent connectivity, and *Centrality* < 0.10 *signifies low influence*, indicating minimal collaboration links. Table VIII shows the details of all categories. VOSViewer employs three key parameters for the analysis, such as Documents, Citations, and TLS. Fig. 11 offers a detailed examination of the document distribution across various defense categories, the Citation and TLS metrics provide deeper insights into the impact and collaboration dynamics. Citations represent the scholarly influence of a country, and TLS reflects the strength and frequency of collaborative relationships between countries. The color gradient bar at the bottom of the image represents the years of activity from 2019 to 2024, with the following color mapping: *Blue (2019), Light Blue (2020), Blue-Green (2021), Green (2022), Greenish-Yellow (2023), and Yellow (2024)*, indicating the chronological progression of country collaborations.

a) Country-Level Analysis (Basic Overview): As publication trends in the Scopus database stabilized around 2019 or slightly earlier, the analysis period for the study was defined from 2019 to 2024. Fig. 12 presents a detailed breakdown of defense-related publications across four key domains—Ground, Aerial, Naval, and Civil Defense from 2019 to 2024 for the top five contributing countries. The data highlights both the annual volume of publications and each country's share in the overall research output, emphasising the active and sustained contributions of *China, India, the United States, and the United Kingdom* across all defense categories. In the domain of ground defense, China holds the largest share, with 118 research articles, accounting for 18.41% of the total publications. The United States and India demonstrate nearly equal contributions, with 156 and 155 publications, respectively, representing approximately 8% each. South Korea and the United Kingdom also show notable involvement, contributing between 4–5% of the global research output in this category. Publications in aerial defense are comparatively fewer than in other defense domains. China leads in the category with 61 research articles, representing 3.5% of the total share. The United States ranks second, contributing 23 publications, which account for approximately 1.30% of the total. The United Kingdom, Turkey, and India also participate in aerial defense research; however, their contributions remain below 1% of the total output. In naval defense, China once again dominates global research contributions, with 338 publications, representing approximately 19% of the total share. The remaining four countries each have fewer than 100 publications. India and the United States exhibit equal contributions,

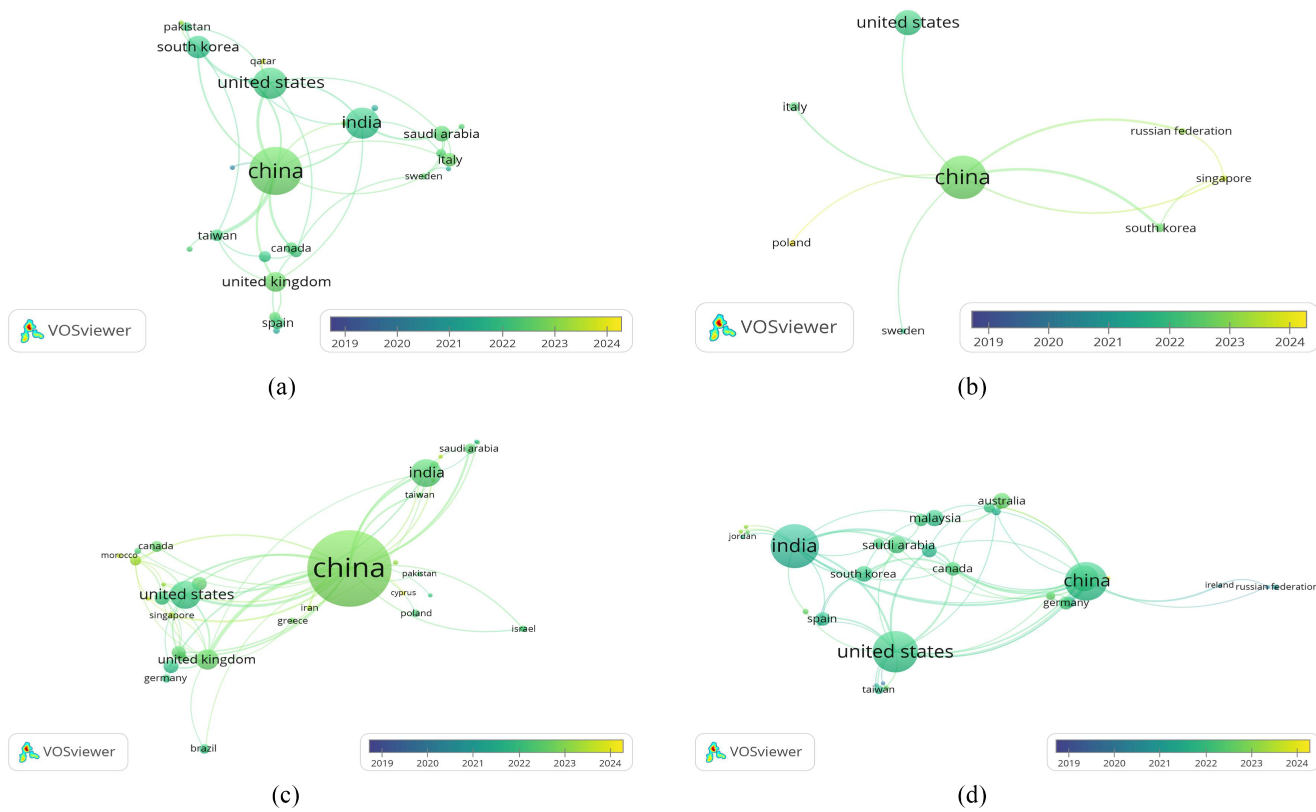

Fig. 11. International contribution analysis across different defense sectors. (a) Country collaboration in ground defense. (b) Country collaboration in aerial defense. (c) Country collaboration in naval defense. (d) Country collaboration in civil defense.

with 72 papers each, accounting for around 4% of the total. The United Kingdom and Italy rank fourth and fifth, with 46 and 25 publications, corresponding to 2.59% and 1.41% of the global naval defense research output, respectively. In civil defense, India leads with the highest number of publications, totalling 94 papers, which accounts for approximately 5.29% of the overall defense research output. The United States and China follow, ranking second and third, with 85 and 77 publications, representing 4.79% and 4.34% of the total share, respectively. The United Kingdom and Saudi Arabia contribute a smaller portion, with less than 3% each.

1) Ground Defense: In the analysis of ground defense research, *94 countries* T_C were examined. Among all, *88 countries* (S_C) met the threshold criteria, which M_D *and* M_C *both set to 1*. The resulting VOSViewer visualization, depicted in Fig. 11(a) shows strong international collaboration in Ground Defense, with countries closely connected through shared research. Figure reflects active knowledge sharing and joint efforts to advance defense technologies. The network with *47 interconnected links and 69 TLS*. The color gradient bar at the bottom of the image indicates the years of activity from *2019 (blue) to 2024 (yellow)*.

1) *2023–2024: Saudi Arabia, China, Italy, Qatar United Kingdom, and Sweden* are the most recent contributors, indicated by the *yellowish-green to yellow color*.
2) *2021–2022: India, United States, South Korea, Pakistan* show collaboration across multiple years, with the colors ranging from green to light green.

China serves as the central hub of ground defense collaboration, showcasing strong ties with multiple countries. With *3713 citations, 30 TLS, and a centrality score of 0.30*, country exerts significant influence within the network. *The United States and the United Kingdom* also demonstrate consistent collaboration, maintaining *centrality scores above 0.20 and recording 2524 and 1494 citations*, respectively, indicating the prominent roles.

In contrast, *India and Saudi Arabia* exhibit moderate influence and connectivity, with *centrality values below 0.20*, reflecting the lesser but still notable contributions to the ground defense collaboration network.

2) Aerial Defense: The analysis of aerial defense research involved the examination of *23 countries* T_C. Out of all, *21 countries* (S_C) satisfied the threshold criteria, with *both* M_D *and* M_C *set to 1*. The resulting VOSViewer visualization, presented in Fig. 11(b) shows the international research network in Aerial Defense, highlighting frequent cross-country collaborations. The structure suggests growing global interest, shared technological goals, and joint contributions to the advancement of aerial defense systems. The network with *8 interconnected links* and *a TLS of 17*. The color gradient bar at the bottom of the image represents the years of activity, ranging from 2019 (blue) to 2024 (yellow), depicting the chronological progression of country collaborations.

1) *2023–2024: Russian Federation, Singapore, and Poland* are the most recent contributors, indicated by the yellowish-green color to yellow color.

	Country	2019	2020	2021	2022	2023	2024	2019-2024	2019	2020	2021	2022	2023	2024	2019-2024
		Number of Publications in Scopus							Share in Total publication output						
	Total	170	174	198	302	392	540	1776	100%	100%	100%	100%	100%	100%	100%
Ground Defense	China	22	21	31	55	80	118	327	12.94%	12.07%	15.66%	18.21%	20.41%	21.85%	18.41%
	United States	17	19	19	25	33	43	156	10.00%	10.92%	9.60%	8.28%	8.42%	7.96%	8.78%
	India	33	8	11	29	38	36	155	19.41%	4.60%	5.56%	9.60%	9.69%	6.67%	8.73%
	South Korea	9	12	13	18	22	15	89	5.29%	6.90%	6.57%	5.96%	5.61%	2.78%	5.01%
	United Kingdom	4	7	8	12	15	30	76	2.35%	4.02%	4.04%	3.97%	3.83%	5.56%	4.28%
Aerial Defense	China		6	9	11	12	23	61		3.45%	4.55%	3.64%	3.06%	4.26%	3.43%
	United States	2		3	10	6	2	23	1.18%		1.52%	3.31%	1.53%	0.37%	1.30%
	United Kingdom				2		6	8				0.66%		1.11%	0.45%
	Turkey			1		2	2	5			0.51%		0.51%	0.37%	0.28%
	India		1		1		2	4		0.57%		0.33%		0.37%	0.23%
Naval Defense	China	17	19	30	53	84	135	338	10.00%	10.92%	15.15%	17.55%	21.43%	25.00%	19.03%
	India	8	8	4	11	16	25	72	4.71%	4.60%	2.02%	3.64%	4.08%	4.63%	4.05%
	United States	19	18	10	8	5	12	72	11.18%	10.34%	5.05%	2.65%	1.28%	2.22%	4.05%
	United Kingdom	1	5	7	8	7	18	46	0.59%	2.87%	3.54%	2.65%	1.79%	3.33%	2.59%
	Italy	1	4	4	5	6	5	25	0.59%	2.30%	2.02%	1.66%	1.53%	0.93%	1.41%
Civil Defense	India	23	9	10	14	24	14	94	13.53%	5.17%	5.05%	4.64%	6.12%	2.59%	5.29%
	United States	6	15	13	15	13	23	85	3.53%	8.62%	6.57%	4.97%	3.32%	4.26%	4.79%
	China	6	14	12	14	14	17	77	3.53%	8.05%	6.06%	4.64%	3.57%	3.15%	4.34%
	United Kingdom	2	5	10	5	8	11	41	1.18%	2.87%	5.05%	1.66%	2.04%	2.04%	2.31%
	Saudi Arabia		3	3	6	7	3	22		1.72%	1.52%	1.99%	1.79%	0.56%	1.24%

Fig. 12. Analysis covers the number of defense-related publications across all categories in the top 5 countries from 2019 to 2024, including all document types and sources indexed in Scopus.

2) *2021–2022: United States, South Korea, Italy, and China* show collaboration during the period, represented by light green to green nodes.

China stands out as the central hub of aerial defense collaboration, exhibiting dominant influence with 5478 citations, *104 TLS, and a high centrality score of 0.54*, making China the most influential player in the network. The *United States* follows as a significant contributor, with *1228 citations, 32 TLS*, and a *centrality of 0.20*, indicating notable but comparatively lower influence. In contrast, *India, the United Kingdom, and Australia* demonstrate moderate connectivity, with *link strengths below 1000* and *centrality values under 0.20*, highlighting the lesser impact in the collaborative network. The *TLS values, all below 25*, further emphasize the limited collaborative strength in aerial defense partnerships.

3) Naval Defense: The naval defense research analysis examined *77 countries* T_C, of which *70 countries* (S_C) met the threshold criteria, with *both the* M_D *and* M_C *set to 1*. The resulting VOSViewer visualization, shown in Fig. 11(c) illustrates the international collaboration network in Naval Defense. The visual shows moderate connectivity among countries, indicating selective partnerships and focused research efforts. The network consists of *77 interconnected links* with a *TLS of 17*.

1) *2023–2024: Morocco, Canada, China, Iran, Singapore, and Cyprus* are the most recent contributors, indicated by the yellowish-green, yellow colors.
2) *2021–2022: China, United Kingdom, India and Saudi Arabia* show collaboration during the period, represented by light green to green nodes.

China emerges as the leading hub in aerial defense collaboration, *with 5478 citations, 104 TLS*, and *a centrality of 0.54*, highlighting China strong influence and extensive network reach. The *United States* follows with *1228 citations, 32 TLS, and a centrality of 0.20*, indicating moderate but notable connectivity. In contrast, *India, the United Kingdom, and Australia* exhibit less prominent roles, with *centrality below 0.20*, *TLS values of 22, 38, and 12, and citations under 1000*, reflecting their moderate influence and limited collaboration in the aerial defense domain.

4) Civil Defense: The civil defense research analysis examined *80 countries* T_C, of which *80 countries* (S_C) met the threshold criteria, with *both the* M_D *and* M_C *set to 1*. The VOSViewer visualization in Fig. 11(d) shows a well-distributed international collaboration network in Civil Defense, reflecting broad participation, consistent knowledge exchange, and collective efforts to enhance resilience and emergency preparedness. The network comprises *83 interconnected links* with a *TLS of 121*, reflecting the extent of international collaboration in civil defense research.

1) 2023–2024: *China, Australia, and Saudi Arabia* exhibit consistent collaboration the most recent contributors, indicated by the *yellowish-green, yellow color*.
2) 2021–2022: The *United States, South Korea, Malaysia, and India* were actively involved during the earlier years, depicted by *light green to green nodes*.

The United States demonstrates a strong influence in the collaboration network with *1285 citations*, *27 TLS*, and a *centrality of 0.20*, indicating moderate but notable connectivity. *China*, despite having *fewer citations (1034)*, holds the *highest centrality at 0.25*, highlighting China strategic importance. *India* shows moderate influence, with *1107 citations*, *29 TLS*, and *a centrality of 0.15*, reflecting consistent but less dominant connectivity. In contrast, *Canada and Pakistan* play minor roles, with *fewer than 500 citations*, *TLS below 25*, and *low centrality values of 0.09 and 0.17*, respectively, indicating limited impact in the network.

F. Key Thematic Areas in Defense

A focused analysis across ground, aerial, and naval defense domains identifies several dominant thematic areas shaping military innovation. AI, ML, and Deep Learning are central to modern defense systems, enabling autonomous threat detection, fault diagnosis, and predictive maintenance. Autonomous platforms such as autonomous aerial, underwater, and ground vehicles are being developed for intelligent navigation, reconnaissance, and mission execution. Cybersecurity and resilient communication infrastructures, including blockchain, 5G, and underwater wireless networks, are critical for secure data transmission in network-centric warfare. DTT and simulations support real-time modeling, mission rehearsal, and system optimization for operational readiness. Situational awareness and surveillance are enhanced through satellite imagery, sensor fusion, and AI-driven object tracking for superior tactical intelligence. Efforts in energy and thermal management aim to support sustainable operations in extreme defense environments. Human-centric defense systems, including soldier health monitoring and human-machine interfaces, strengthen decision support and combat effectiveness. Lastly, intelligent combat modeling and decision-making systems, leveraging reinforcement learning and expert systems, enable adaptive responses in complex battlefield scenarios.

V. Systematic Literature Review

The section examines recent literature across all defense categories, focusing on the integration of emerging technologies such as AI, ML, IoT, DT, and 5G/6G networks. The objective is to deliver a comprehensive overview of current advancements, outline key research challenges, and emphasize future research directions essential for the progression of modern defense strategies.

Ground Defense: Fossacece and Young [12] and Johnson [66] both emphasized the growing role of AI in ground defense operations. Johnson highlights the risks of automation bias caused by AI anthropomorphism in HMIs, stressing the need for human-in-the-loop oversight to ensure ethical and reliable AI decisions. Complementing this, Fossacece et al. focus on the U.S. Army's AI/ML research, which enhances the autonomy of ground robots using nonparametric learning algorithms for real-time scene understanding. Their work also includes human-assisted learning and natural language communication to improve AI adaptability and decision-making in uncertain environments. Szabadföldi [67] and Rettore et al. [68] both explored AI-driven technological advancements in defense operations. Szabadfoldi et al. discuss AI's transformative impact on military operations by enhancing intelligence analysis, decision-making, and autonomous systems. Authors highlight AI's role in tackling Big Data challenges while addressing reliability issues like artificial stupidity. In parallel, Rettore et al. focus on the Military IoT (MIoT), which improves situational awareness and decision-making by interconnecting heterogeneous devices. Authors emphasize the role of AI and data fusion techniques in enhancing MIoT efficiency despite challenges like mobility and wireless interference. Malik et al. [69] highlighted the integration of advanced networking technologies in military operations. Malik et al. focus on 5G's role in enhancing communication capabilities with features like private networks, blockchain security, and IoT integration for real-time data fusion and drone control.

Aerial Defense: Keneni et al. [38], Ernest et al. [70], and Yu et al. [71] focused on AI-powered UAV navigation and decision-making for aerial defense. Ernest et al. introduce ALPHA, using genetic fuzzy tree (GFT) and fuzzy logic for real-time combat tactics like flanking and evasion. Yu et al. provide a comprehensive review of UAV cyberattacks, categorizing threats into UAV-to-UAV coordination, UAV-to-command center links, and onboard systems, while exploring anomaly detection, threat avoidance, and future defenses like federated learning, adversarial AI, and blockchain-based authentication. Keneni et al. present an XAI model using Sugeno-type FIS to predict weather and enemy impact on UAV missions, achieving high accuracy (RMSE < 0.05), thereby demonstrating significant value for threat response scenarios. Prabagar et al. [72] and Beltan et al. [73] emphasized AI-driven UAV efficiency and robustness in dynamic environments. Prabagar uses deep reinforcement learning with Proximal Policy Optimization to enhance UAV accuracy, adaptability, and resilience against wind disturbances. Real-world tests show precision and quick response times, with future directions in ML techniques, safety protocols, and urban integration. Beltan et al. introduce Flighter, a DFL-based framework for reconnaissance missions, combining situational awareness with defense modules to counter adversarial attacks. Detection accuracy is enhanced through the use of SAR data, while the impact of attacks is minimized, resulting in increased effectiveness for aerial defense operations.

Naval Defense: Recent studies have introduced ML-driven frameworks to enhance naval cybersecurity, predictive maintenance, and autonomous operations. Lan et al. [74] and Kumar et al. [75] proposed adaptive defense models for cyber–physical Systems and IoT, ensuring digital sovereignty through ML-infused analytics. Lan et al. AFL-VA-ISN model detects cyberattacks with a 98.5% accuracy and reduces network latency by 11.4%, while Kumar et al. Neoteric Threat Intelligence framework strengthens 6G and Beyond network security, even in resource-constrained IoT environments. For predictive maintenance and automation, Krishna et al. [76] explored ML-based fault prediction in naval fleets, boosting fleet readiness and minimizing downtime through AI-driven route optimization. Lim et al. [77] enhanced mine detection accuracy in Manned and autonomous Teaming systems by augmenting Side Scan Sonar data with the YOLOv8 object detection model. In maritime communication, Kim et al. [78] introduced an AI-powered voice communication system for Maritime Autonomous Surface Ships, integrating Speech-to-Text, Text-to-Speech, and Large Language Models to enable real-time, standardized maritime responses and improve situational awareness.

Civil Defense: Blount [79], Adams [80], and Dubravova et al. [81] examined the growing influence of AI in law enforcement, highlighting the impact on policing practices, officer discretion, and civil defense strategies. Blount underscores how AI-driven predictive policing risks eroding due process protections, promoting indirect discrimination, and undermining the presumption of innocence due to flawed risk assessments. Similarly, Adams reveals, AI powered monitoring through Body-Worn Cameras reduces perceived fairness and lowers officer trust and morale, affecting transparency and operational efficiency. Dubravova et al. emphasize how AI enhances crime prevention, predictive analytics, and administrative efficiency, but also raise concerns over privacy and ethical implications. Ardabili et al. [82] further explored AI in public safety through Smart Video Surveillance systems, integrating computer vision, statistical analytics, and cloud services. To address privacy concerns, adopt pose-based algorithms, which reduce accuracy in anomaly and action detection by 23.76% and 12.93%, respectively. Despite achieving 17.8 FPS in extreme video scenes, a latency of 36.1 s restricts real-time effectiveness. Collective findings indicate that AI enhances efficiency in policing and surveillance while presenting substantial concerns related to privacy, fairness, and due process. The findings call for improved accuracy, reduced latency, and stronger accountability measures to ensure the ethical and effective deployment of AI in civil defense and law enforcement.

Overall, the scientometric analysis demonstrates how 4G, 5G, 6G, IoT, ML, and AI are transforming defense operations across all domains. From UAV navigation and naval cybersecurity to ground combat decision-making and civil surveillance, the technologies enhance autonomous responses, operational efficiency, and tactical precision, shaping the future of modern defense strategies.

VI. Discussion

The section aims to address the previously defined research questions through a scientometric and systematic literature analysis of defense related studies from the Scopus dataset. The analysis is conducted across four key defense categories: *Ground, Aerial, Naval, and Civil*, offering a comprehensive examination of research trends, collaboration networks, and technological advancements.

A. Key Findings and Trends

The scientometric analysis reveals significant growth in defense related research from 2019 to 2024, driven by the increasing adoption of emerging technologies such as AI, ML, IoT, DT, and 5G/6G networks.

1) *Overall Research Growth (2019–2024):* The RQ1 is addressed here, the publication trends indicate a steady rise in scholarly output, with a particular focus on ground and naval defense, which collectively account for the largest share of publications. In comparison, aerial defense demonstrates moderate growth, while civil defense, though relatively less explored, is gradually gaining attention.
2) *Ground Defense:* Research emphasizes AI-powered surveillance systems for real-time threat detection, autonomous robotics for battlefield operations, battlefield analytics for decision support, and threat detection.
3) *Naval defense* highlights cybersecurity frameworks to protect maritime networks, predictive maintenance models powered by AI to reduce downtime, and autonomous maritime systems for patrolling and defense operations.
4) *Aerial Defense:* Largely driven by UAV technologies for aerial surveillance and reconnaissance, autonomous combat systems, and AI-based navigation.
5) *Civil defense* is gaining traction with studies focusing on AI-driven law enforcement for predictive policing, public safety systems with real-time surveillance, and smart surveillance leveraging facial recognition and anomaly detection. Research on AI-powered crowd monitoring, predictive policing, and emergency response systems showcases the increasing integration of smart technologies in civil security.

B. Comparison With Existing Research

To validate the scientometric findings and highlight the novelty and relevance of the current study,a comparative review was conducted against selected studies. The comparison was based on methodological depth, domain coverage, and thematic resolution. While prior works such as Kaur and Saini (2024) [32] and Videras Razzaq and Shah (2025) [83] focused on specific domains like cybersecurity or cyber–physical systems using basic co-authorship or co-citation metrics, and lacked multidomain defense stratification. More recent efforts like Cecere and Greco (2025) [84] offered broader coverage but emphasized literature trends without assessing cluster quality or thematic integration. In contrast, the present study spans four key defense domains—ground, naval, aerial, and civil and applies advanced CiteSpace-based DCA, modularity, and silhouette metrics to extract high-quality thematic clusters and provides a comprehensive view of emerging technologies such as AI, DT, 5G/6G, and autonomous systems in defense, addressing gaps in depth, domain breadth, and strategic foresight found in earlier studies. Table IX presents a comparative analysis of bibliometric and scientometric studies based on four key parameters, such as approach, methods, key findings, and limitations.

C. Leading Contributors and Research Impact

1) RQ2 reveals AI, ML, UAV, cybersecurity, DT, and IoT are the most frequently co-occurring keywords, highlighting the technological convergence shaping modern defense research. The prominence of AI and ML reflects the widespread application in autonomous systems, surveillance, and decision-making, while UAV and cybersecurity indicate the growing emphasis on aerial operations and network protection. The frequent occurrence of DT and IoT underscores the increasing reliance on digital replicas and interconnected devices for real-time monitoring and predictive analytics.
2) RQ3 reveals Johnson (2024) and Fossacece (2018) as key references in ground defense, frequently cited for contributions to AI-powered surveillance and robotics-driven operations. In naval defense, Lan et al. (2024) and Kumar et al. (2025) are highly co-cited for work

TABLE IX
COMPARATIVE TABLE OF BIBLIOMETRIC AND SCIENTOMETRIC STUDIES

Reference	Year	Approach	Methods	Key Findings	Limitations
[37]	2025	Literature & Bibliometric Review	Scopus rankings, keyword analysis	Identified top 5 global journals; analyzed top 10 Russian periodicals; highlighted issues in Russian journals (low diversity, high self-citation).	Need for increased international author diversity and editorial reforms in Russian journals.
[32]	2024	Scientometric Analysis	Co-Authorship (CoNA), Co-Citation (CcNA)	Identified five thematic clusters (ML, DL, etc.); observed global collaboration and keyword co-occurrence; focused on AI integration in CPPA over a decade.	Excludes unpublished/ongoing studies; suggests using alternative data sources and tools beyond VOSviewer.
[35]	2024	Bibliometric Analysis	Temporal trends, term analysis	Observed growth in USV research (2000–2023); identified top authors, institutions, journals; hotspots include autonomy, sensors, endurance, eco-friendly practices; proposed six future directions.	Lack of comprehensive reviews on recent USV technologies and trends.
[85]	2022	Scientometric	HDBSCAN, Co-authorship, Source Coupling	Mapped institutional collaboration patterns; identified author dominance clusters; highlighted central institutions.	Absence of defense-specific insights; lacks co-citation and thematic clustering; not domain-specific.
[86]	2016	Bibliometric Analysis	Degree of Collaboration, Collaborative Coefficient	Analyzed 783 publications (2005–2014); 77.91% scholarly articles; peak collaboration in 2008 and 2012 (0.913); India contributed 77.73%.	Limited international diversity in authorship; suggests enhancing global participation and collaboration.
[84]	2025	Scientometric and Systematic Literature Analysis	Co-Citation Analysis (DCA), Keyword Co-Occurrence, Country Collaboration Network, Cluster Analysis (Modularity Q and Silhouette S), Thematic Trend Mapping using CiteSpace	Provided an in-depth, multi-domain exploration of emerging technologies across Ground, Naval, Aerial, and Civil Defense sectors; employed rigorous DCA and multi-level cluster validation, identified global leaders, key authors, and advanced tech adoption trends (AI, DT, 5G/6G); formulated a forward-looking roadmap addressing real-world deployment challenges.	Despite the comprehensive analytical scope, the study primarily focuses on scholarly literature. Real-world validation of technology integration such as 6G or DT in operational defense remains a future direction.
[83]	2025	Scientometric and Bibliometric Analysis	Co-authorship mapping, Citation analysis, Country and Institutional Productivity, Keyword Co-occurrence (VOSviewer)	Provided an in-depth bibliometric study on the integration of ML/DL in cybersecurity; identified 3712 documents across 114 countries, with Saudi Arabia as the top contributor. Highlighted influential authors, top institutions, key sources (IEEE, Sensors), and emerging research themes like adversarial learning, AI, and blockchain-based security.	Lacks domain stratification and co-citation analysis (e.g., DCA); focuses on document volume and network structure without cluster quality measures or theme evolution; does not propose a multi-dimensional framework for future cybersecurity policy or standardisation.
Present Study	2025	Scientometric and Systematic Literature Analysis	Co-Citation Analysis (DCA), Keyword Co-Occurrence, Country Collaboration Network, Cluster Analysis (S/Q), Thematic Trend Mapping	Delivered a domain-wide scientometric exploration across Ground, Naval, Aerial, and Civil Defense; utilized multi-layered CiteSpace analytics to identify top authors, institutions, global collaborations, emerging technologies (AI, ML, DT, UAVs, 5G/6G); proposed a future-ready research agenda for defense innovation.	Previous studies lacked defense-specific focus, domain stratification, and integrated tech foresight; this study bridges those gaps with comprehensive clustering, advanced metrics, and interdisciplinary insights, setting a new benchmark for scientometric rigor in defense research.

on maritime cybersecurity and predictive maintenance. For aerial defense, Ernest et al. (2016) and Keneni et al. (2019) are prominent for their focus on UAV navigation and AI-based combat strategies. In civil defense, Blount (2024) and Adams (2025) are influential for research on AI-powered law enforcement and public safety. The frequent co-citation of these studies underscores significant influence on defense research, shaping technological advancements across all sectors.

3) RQ4 China as the dominant force in defense research, leading in ground and naval defense with the highest CC, centrality, and TLS, reflecting China strong influence and extensive collaborations. The United States follows closely, excelling in aerial and civil defense, particularly in UAV technologies and AI driven public safety systems. India and the United Kingdom also demonstrate significant contributions, notably in civil and naval defense, while Saudi Arabia, South Korea, and Australia emerge as growing players, focusing on UAV operations and maritime security. The global collaboration network reveals robust partnerships, with China and the U.S. forming the most influential alliance, while India, the U.K., and Australia exhibit increasing bilateral cooperation, driving advancements in defense technology.
4) The scientometric analysis reveals a profound real-world impact, with defense research significantly shaping industry practices and operational strategies. The adoption of AI, ML, and DTT in battlefield surveillance, maritime cybersecurity, and UAV navigation is transforming defense operations, enhancing situational awareness, automation, and decision-making. The increasing focus on predictive analytics, autonomous systems, and real-time monitoring highlights the field's growing practical relevance in modern defense strategies.

D. Emerging Technology

1) *5G/6G Networks:* Emerging as a crucial enabler of faster data transmission, enhanced connectivity, and low-latency communication, particularly for UAVs and autonomous maritime systems.
2) *IoT and DT Integration:* Increasingly applied in real-time monitoring and situational awareness, particularly in naval and civil defense. The use of MIoT networks improves communication efficiency and data fusion.
3) *AI and ML Algorithms:* Widely used for predictive maintenance, threat detection, and autonomous decision-making. Studies demonstrate high accuracy rates and operational efficiency gains using techniques such as Deep Reinforcement Learning and YOLOv8 object detection models.

E. Research Gaps, Challenges, and Future Insights

To address RQ5, which explores emerging trends and future insights in defense research, the scientometric analysis highlights several key developments shaping the field. The increasing adoption of AI-powered autonomous systems is transforming battlefield operations, with advancements in UAV navigation, robotics, and autonomous combat systems enhancing operational capabilities. DT technology is gaining traction, enabling real-time simulation, predictive maintenance, and fleet management, thereby boosting efficiency and reducing downtime. The integration of 5G/6G networks is revolutionizing defense communication by facilitating ultrafast, low-latency data exchange, improving situational awareness, and decision-making. Future research will prioritize cross-domain interoperability, multidomain operations, and ethical AI deployment, ensuring the development of trustworthy, resilient, and adaptive defense technologies. The following points explain the research gaps along with the challenges and future insights, providing a comprehensive overview of the field's evolving landscape.

1) *Limited Real World Implementation and Validation:*
   a) *Gap:* Despite the growing adoption of AI and DT in defense research, real-world implementation and validation remain limited. Many studies focus on theoretical models and simulated environments rather than practical deployment in operational defense settings.

b) *Challenge:* The lack of real-world testing raises concerns about the reliability, scalability, and robustness of these technologies when applied to dynamic and unpredictable defense scenarios.

c) *Future Direction:* Greater emphasis on field trials, pilot projects, and operational testing is required to bridge the gap between research and practical deployment.

2) *Data Security, Privacy, and Integrity Concern:*

a) *Gap:* With the increasing adoption of IoT and AI-powered surveillance, defense systems are generating massive volumes of sensitive data. However, ensuring data security and privacy in large-scale networks remains a major challenge.

b) *Challenge:* The lack of standardized security protocols makes defense networks vulnerable to cyberattacks, data breaches, and unauthorized access.

c) *Future Direction:* Blockchain-based encryption, zero-trust architectures, and AI-powered cybersecurity frameworks are essential to enhance data protection and prevent cyber vulnerabilities.

3) *Scalability and Infrastructure Limitations:*

a) *Gap:* Deploying large-scale DT, AI, and 6G powered defense systems requires robust infrastructure and computational resources. However, many defense organizations face scalability challenges due to hardware limitations and network constraints.

b) *Challenge:* The limited scalability of defense networks restricts the ability to handle massive data volumes, ensure seamless connectivity, and support real-time operations.

c) *Future Direction:* Investing in high-performance computing infrastructure, cloud-based defense solutions, and 6G-enabled networks will enhance the scalability and efficiency of defense systems.

4) *Interoperability and Standardization Issues:*

a) *Gap:* The fragmented technological landscape across different defense sectors (ground, aerial, naval, and civil) creates significant interoperability challenges, making seamless system integration difficult.

b) *Challenge:* The lack of standardized data formats, communication protocols, and frameworks hinders the synchronization of diverse defense technologies, reducing operational efficiency.

c) *Future Direction:* Establishing unified communication protocols, cross-domain compatibility standards, and interoperable frameworks is essential to enhance coordination, streamline data exchange, and improve overall efficiency in defense operations.

VII. Conclusion

The study provides a comprehensive scientometric and systematic analysis of emerging technologies in the defense sector from 2019 to 2024, highlighting the application across *Ground, Aerial, Naval, and Civil domains*. By leveraging tools like CiteSpace and VOSViewer, the research maps key developments, collaborations, and thematic trends. The analysis demonstrates a significant rise in defense research over the years, with a notable surge between 2019 and 2024, driven by the accelerated adoption of AI-powered autonomous vehicles, predictive analytics, and real-time situational awareness systems. The DCA and keyword co-occurrence networks highlight key research clusters, focusing on areas such as cybersecurity, autonomous defense systems, battlefield simulation, surveillance, and maritime security. The work serves as a foundational reference for researchers and policymakers exploring technology-driven defense innovation. However, the study has a few limitations.

1) *Limited Temporal Coverage:* The analysis is confined to the period between 2019 and 2025, potentially excluding foundational or seminal research conducted prior to this timeframe.
2) *Language Restriction:* Only English-language publications are included in the dataset, which may overlook important studies published in other languages.
3) *Absence of Empirical Validation:* The findings are derived exclusively from academic literature without cross-verification through real-world deployment data, field trials, or operational case studies.
4) *Database Limitation:* The study relies solely on data extracted from Scopus databases, which may omit relevant articles published in other sources.

To address existing gaps, future research should incorporate implementation-based evaluations, cross-domain field testing, and real-time data to assess the practical impact and scalability of emerging technologies in active defense scenarios. Additionally, future studies should broaden linguistic and temporal coverage to ensure more comprehensive and globally representative insights. Moreover, researchers should explore how technologies like AI, IoT, and DT can be integrated across multiple defense domains, enabling unified situational awareness and joint operational effectiveness.

References

[1] A. Jyothi, A. Shankar, J. R. A. Narayan, K. Bhavya, and S. Y. MM, "AI & ML methodologies in navy & their systems," in *Proc. IEEE Int. Conf. Evol. Algorithms Soft Comput. Techn. (EASCT)*, 2023, pp. 1–6.

[2] C. Benzaïd and T. Taleb, "AI for beyond 5G networks: A cyber-security defense or offense enabler?" *IEEE Netw.*, vol. 34, no. 6, pp. 140–147, Nov./Dec. 2020.

[3] W. J. Perry, "Military technology: An historical perspective," *Technol. Soc.*, vol. 26, nos. 2–3, pp. 235–243, 2004.

[4] A. B. Rashid, A. K. Kausik, A. A. H. Sunny, and M. H. Bappy, "Artificial intelligence in the military: An overview of the capabilities, applications, and challenges," *Int. J. Intell. Syst.*, vol. 2023, no. 1, 2023, Art. no. 8676366.

[5] E. J. A. Suárez and V. M. Baeza, "Evaluating the role of machine learning in defense applications and industry," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 4, pp. 1557–1569, 2023. [Online]. Available: https://www.mdpi.com/2504-4990/5/4/78

[6] A. Petrovski, M. Radovanović, and A. Behlić, "Application of drones with artificial intelligence for military purposes," in *Proc. 10th Int. Sci. Conf. Defensive Technol. (OTEH)*, 2022, pp. 92–100.

[7] C. Rikap, "The U.S. National Security State and big tech: Frenemy relations and innovation planning in turbulent times," *Rev. Keynesian Econ.*, vol. 12, no. 3, pp. 348–364, 2024.

[8] P. Svenmarck, L. Luotsinen, M. Nilsson, and J. Schubert, "Possibilities and challenges for artificial intelligence in military applications," in *Proc. NATO Big Data Artif. Intell. Mil. Decis. Making Specialists' Meeting*, vol. 1, 2018, p. 12.

[9] H. Saarnisaari, M. Höyhtyä, H. Rantanen, and J. Mäkelä, "Military communications in the 6G era: Finnish perspective," in *Proc. IEEE Mil. Commun. Conf. (MILCOM)*, 2024, pp. 135–140.

[10] A. F. Mendi, T. Erol, and D. Doğan, "Digital twin in the military field," *IEEE Internet Comput.*, vol. 26, no. 5, pp. 33–40, Sep./Oct. 2022.

[11] T. Zheng, M. Liu, D. Puthal, P. Yi, Y. Wu, and X. He, "Smart grid: Cyber attacks, critical defense approaches, and digital twin," 2022, *arXiv:2205.11783*.

[12] J. M. Fossaceca and S. H. Young, "Artificial intelligence and machine learning for future army applications," in *Proc. SPIE*, 2018, pp. 8–25.

[13] J. Wu, Y. Yang, X. Cheng, H. Zuo, and Z. Cheng, "The development of digital twin technology review," in *Proc. IEEE Chin. Autom. Congr. (CAC)*, 2020, pp. 4901–4906.

[14] G. Lazaroiu et al., "Digital twin-based cyber-physical manufacturing systems, extended reality metaverse enterprise and production management algorithms, and Internet of Things financial and labor market technologies in generative artificial intelligence economics," *Oeconomia Copernicana*, vol. 15, no. 3, pp. 837–870, 2024.

[15] G. Lazaroiu et al., "Cognitive digital twin-based internet of robotic things, multi-sensory extended reality and simulation modeling technologies, and generative artificial intelligence and cyber–physical manufacturing systems in the immersive industrial metaverse," *Equilibrium. Quart. J. Econ. Econ. Policy*, vol. 19, no. 3, pp. 719–748, 2024.

[16] C. Gamage, R. Dinalankara, J. Samarabandu, and A. Subasinghe, "A comprehensive survey on the applications of machine learning techniques on maritime surveillance to detect abnormal maritime vessel behaviors," *WMU J. Maritime Affairs*, vol. 22, no. 4, pp. 447–477, 2023.

[17] C.-E. Lee, J. Baek, J. Son, and Y.-G. Ha, "Deep AI military staff: Cooperative battlefield situation awareness for Commander's decision making," *J. Supercomput.*, vol. 79, no. 6, pp. 6040–6069, 2023.

[18] I. Durlik, T. Miller, D. Cembrowska-Lech, A. Krzemińska, E. Złoczowska, and A. Nowak, "Navigating the sea of data: A comprehensive review on data analysis in maritime IoT applications," *Appl. Sci.*, vol. 13, no. 17, p. 9742, 2023.

[19] N. Telagam, N. Kandasamy, A. K. Manoharan, P. Anandhi, and R. Atchudan, "Beyond 5G: Exploring key enabling technologies, use cases, and future prospects of 6 G communication," *Nano Commun. Netw.*, vol. 43, Mar. 2025, Art. no. 100560. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1878778924000668

[20] S. Dinh-Van, T. M. Hoang, B. B. Cebecioglu, D. S. Fowler, Y. K. Mo, and M. D. Higgins, "A defensive strategy against beam training attack in 5G mmWave networks for manufacturing," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2204–2217, 2023.

[21] D. J. Fagnant and K. Kockelman, "Preparing a nation for autonomous vehicles: Opportunities, barriers and policy recommendations," *Transp. Res. A Policy Pract.*, vol. 77, pp. 167–181, 2015.

[22] P. A. Ganie, A. Khatei, R. Posti, M. J. Sidiq, and P. K. Pandey, "Unmanned aerial vehicles in fisheries and aquaculture: A comprehensive overview," *Environ. Monitor. Assess.*, vol. 197, no. 5, p. 503, 2025.

[23] S. Sai, A. Garg, K. Jhawar, V. Chamola, and B. Sikdar, "A comprehensive survey on artificial intelligence for unmanned aerial vehicles," *IEEE Open J. Veh. Technol.*, vol. 4, pp. 713–738, 2023.

[24] U. O. Matthew, J. S. Kazaure, A. Onyebuchi, O. O. Daniel, I. H. Muhammed, and N. U. Okafor, "Artificial intelligence autonomous unmanned aerial vehicle (UAV) system for remote sensing in security surveillance," in *Proc. IEEE 2nd Int. Conf. Cyberspace (CYBER NIGERIA)*, 2021, pp. 1–10.

[25] S. Raaijmakers, "Artificial intelligence for law enforcement: Challenges and opportunities," *IEEE Security Privacy*, vol. 17, no. 5, pp. 74–77, Sep./Oct. 2019.

[26] T. Sushina and A. Sobenin, "Artificial intelligence in the criminal justice system: Leading trends and possibilities," in *Proc. 6th Int. Conf. Soc. Econ. Acad. Leadership (ICSEAL)*, 2020, pp. 432–437. [Online]. Available: https://doi.org/10.2991/assehr.k.200526.062

[27] T. Kliestik et al., "Enterprise generative artificial intelligence technologies, Internet of Things and blockchain-based fintech management, and digital twin industrial metaverse in the cognitive algorithmic economy," *Oeconomia Copernicana*, vol. 15, no. 4, pp. 1183–1221, 2024.

[28] M. R. Brust, G. Danoy, D. H. Stolfi, and P. Bouvry, "Swarm-based counter UAV defense system," *Discover Internet Things*, vol. 1, no. 1, pp. 1–19, 2021.

[29] M. Mozaffari, W. Saad, M. Bennis, and M. Debbah, "Unmanned aerial vehicle with underlaid device-to-device communications: Performance and tradeoffs," *IEEE Trans. Wireless Commun.*, vol. 15, no. 6, pp. 3949–3963, Jun. 2016.

[30] V. P. Karamchedu, "A path from device-to-device to UAV-to-UAV communications," in *Proc. IEEE 92nd Veh. Technol. Conf. (VTC-Fall)*, 2020, pp. 1–5.

[31] I. H. Abdulqadder, S. Zhou, D. Zou, I. T. Aziz, and S. M. A. Akber, "Multi-layered intrusion detection and prevention in the SDN/NFV enabled cloud of 5G networks using AI-based defense mechanisms," *Comput. Netw.*, vol. 179, Oct. 2020, Art. no. 107364.

[32] M. Kaur and M. Saini, "Role of artificial intelligence in the crime prediction and pattern analysis studies published over the last decade: A scientometric analysis," *Artif. Intell. Rev.*, vol. 57, no. 8, p. 202, 2024.

[33] M. V. Rodríguez, S. G. Melgar, A. S. Cordero, and J. M. A. Márquez, "A critical review of unmanned aerial vehicles (UAVs) use in architecture and urbanism: scientometric and bibliometric analysis," *Appl. Sci.*, vol. 11, no. 21, p. 9966, 2021.

[34] J. Wang et al., "Social network and bibliometric analysis of unmanned aerial vehicle remote sensing applications from 2010 to 2021," *Remote Sens.*, vol. 13, no. 15, p. 2912, 2021.

[35] P. Yang, J. Xue, and H. Hu, "A bibliometric analysis and overall review of the new technology and development of unmanned surface vessels," *J. Mar. Sci. Eng.*, vol. 12, no. 1, p. 146, 2024.

[36] A. K. Pandey, A. Chakraborty, and V. Khandal, "Scientometric study of research on AI & ML application in defence technology and military operations," *DESIDOC J. Lib. Inf. Technol.*, vol. 44, no. 2, p. 158, 2024.

[37] D. Taranovskiy and O. Yashnikova, "Review of scientific and technical periodicals on marine navigation equipment engineering," *Gyroscopy Navig.*, vol. 15, no. 1, pp. 99–107, 2024.

[38] B. M. Keneni et al., "Evolving rule-based explainable artificial intelligence for unmanned aerial vehicles," *IEEE Access*, vol. 7, pp. 17001–17016, 2019.

[39] A. A. Alquwayzani and A. A. Albuali, "A systematic literature review of zero trust architecture for military UAV security systems," *IEEE Access*, vol. 12, pp. 176033–176056, 2024.

[40] Y. Yao et al., "Training with product digital twins for autoretail checkout," 2023, *arXiv:2308.09708*.

[41] G. Lăzăroiu, T. Gedeon, D. Szpilko, and K. Halicka, "Digital twin-based alternate ego modeling and simulation: Eva Herzigová as a 3-D MetaHuman avatar," *Eng. Manag. Prod. Services*, vol. 16, no. 3, pp. 1–14, 2024.

[42] M. Kahili-Heede and K. Hillgren, "Colandr," *J. Med. Lib. Assoc. JMLA*, vol. 109, no. 3, p. 523, 2021.

[43] Z. Munn et al., "The development of software to support multiple systematic review types: The Joanna Briggs institute system for the unified management, assessment and review of information (JBI SUMARI)," *JBI Evidence Implement.*, vol. 17, no. 1, pp. 36–43, 2019.

[44] J. T. Minion et al., "PICO portal," *J. Can. Health Lib. Assoc.*, vol. 42, no. 3, p. 181, 2021.

[45] N. Johnson and M. Phillips, "Rayyan for systematic reviews," *J. Electron. Resource Librarianship*, vol. 30, no. 1, pp. 46–48, 2018.

[46] I. J. Saldanha, B. T. Smith, E. Ntzani, J. Jap, E. M. Balk, and J. Lau, "The systematic review data repository (SRDR): Descriptive characteristics of publicly available data and opportunities for research," *Syst. Rev.*, vol. 8, pp. 1–12, Jan. 2019.

[47] B. E. Howard et al., "SWIFT-active screener: Accelerated document screening through active learning and integrated recall estimation," *Environ. Int.*, vol. 138, May 2020, Art. no. 105623.

[48] V. K. Singh, P. Singh, M. Karmakar, J. Leta, and P. Mayr, "The journal coverage of Web of science, Scopus and dimensions: A comparative analysis," *Scientometrics*, vol. 126, pp. 5113–5142, 2021.

[49] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[50] W. Hu and Y. Tan, "Generating adversarial malware examples for black-box attacks based on GAN," in *Proc. Int. Conf. Data Min. Big Data*, 2022, pp. 409–423.

[51] K. T. Giang, S. Song, and S. Jo, "Curvature-guided dynamic scale networks for multi-view stereo," 2021, *arXiv:2112.05999*.

[52] F. Wang, S. Galliani, C. Vogel, P. Speciale, and M. Pollefeys, "PatchmatchNet: Learned multi-view patchmatch stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 14194–14203.

[53] Q. Yang, J. Zhang, G. Shi, J. Hu, and Y. Wu, "Maneuver decision of UAV in short-range air combat based on deep reinforcement learning," *IEEE Access*, vol. 8, pp. 363–378, 2019.

[54] O. Vinyals et al., "Grandmaster level in StarCraft II using multi-agent reinforcement learning," *Nature*, vol. 575, no. 7782, pp. 350–354, 2019.
[55] Z. Yang, D. Zhou, H. Piao, K. Zhang, W. Kong, and Q. Pan, "Evasive maneuver strategy for UCAV in beyond-visual-range air combat based on hierarchical multi-objective evolutionary algorithm," *IEEE Access*, vol. 8, pp. 46605–46623, 2020.
[56] H. Shin, J. Lee, H. Kim, and D. H. Shim, "An autonomous aerial combat framework for two-on-two engagements based on basic fighter maneuvers," *Aerosp. Sci. Technol.*, vol. 72, pp. 305–315, Jan. 2018.
[57] F. Hu, G.-S. Xia, J. Hu, and L. Zhang, "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sens.*, vol. 7, no. 11, pp. 14680–14707, 2015.
[58] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogrammetry remote Sens.*, vol. 145, no. 1, pp. 120–147, 2018.
[59] R. Cui, C. Yang, Y. Li, and S. Sharma, "Adaptive neural network control of AUVs with control input nonlinearities using reinforcement learning," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 6, pp. 1019–1029, Jun. 2017.
[60] S. Fujimoto, H. Hoof, and D. Meger, "Addressing function approximation error in actor–critic methods," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 1587–1596.
[61] C. E. Okereke et al., "An overview of machine learning techniques in local path planning for autonomous underwater vehicles," *IEEE Access*, vol. 11, pp. 24894–24907, 2023.
[62] H. Lang et al., "Multisource heterogeneous transfer learning via feature augmentation for ship classification in SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[63] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
[64] S. Brayne, "Big data surveillance: The case of policing," *Amer. Sociol. Rev.*, vol. 82, no. 5, pp. 977–1008, 2017.
[65] M. T. Bhatti, M. G. Khan, M. Aslam, and M. J. Fiaz, "Weapon detection in real-time CCTV videos using deep learning," *IEEE Access*, vol. 9, pp. 34366–34382, 2021.
[66] J. Johnson, "Finding AI faces in the moon and armies in the clouds: Anthropomorphising artificial intelligence in military human–machine interactions," *Global Soc.*, vol. 38, no. 1, pp. 67–82, 2024.
[67] I. Szabadföldi, "Artificial intelligence in military application—Opportunities and challenges," *Land Forces Acad. Rev.*, vol. 26, no. 2, pp. 157–165, 2021.
[68] P. H. Rettore, J. Mast, T. Aurisch, A. C. Viana, P. Sevenich, and B. P. Santos, "Military IoT from management to perception: Challenges and opportunities across layers," *IEEE Internet Things Mag.*, vol. 8, no. 2, pp. 25–31, Mar. 2025.
[69] M. Malik, A. Kothari, and R. A. Pandhare, "Stand alone or non stand alone 5G tactical edge network architecture for military and use case scenarios," in *Proc. IEEE Int. Conf. Intell. Signal Process. Effective Commun. Technol. (INSPECT)*, 2024, pp. 1–6.
[70] N. Ernest, D. Carroll, C. Schumacher, M. Clark, K. Cohen, and G. Lee, "Genetic fuzzy based artificial intelligence for unmanned combat aerial vehicle control in simulated air combat missions," *J. Defense Manag.*, vol. 6, no. 1, pp. 2167–0374, 2016.
[71] A. Yu, I. Kolotylo, H. A. Hashim, and A. E. Eltoukhy, "Electronic warfare cyberattacks, countermeasures and modern defensive strategies of UAV avionics: A survey," *IEEE Access*, vol. 13, pp. 68660–68681, 2025.
[72] S. Prabagar, A. K. Al-Jiboory, P. S. Nair, P. Mandal, K. M. Garse et al., "Artificial intelligence-based control strategies for unmanned aerial vehicles," in *Proc. 10th IEEE Uttar Pradesh Int. Conf. Elect. Electron. Comput. Eng. (UPCON)*, vol. 10, 2023, pp. 1624–1629.
[73] E. T. M. Beltran, P. M. S. Sanchez, G. Bovet, B. Stiller, G. M. Perez, and A. H. Celdran, "Flighter: Decentralized federated learning and situational awareness for secure military aerial reconnaissance," *IEEE Commun. Mag.*, early access, Mar. 7, 2025, doi: 10.1109/MCOM.003.2400401.
[74] D. Lan, P. Xu, J. Nong, J. Song, and J. Zhao, "Application of artificial intelligence technology in vulnerability analysis of intelligent ship network," *Int. J. Comput. Intell. Syst.*, vol. 17, no. 1, p. 147, 2024.
[75] S. Kumar, S. K. Singh, R. Kumar, C. K. Subba, K. T. Chui, and B. B. Gupta, "Neoteric threat intelligence ensuring digital sovereignty and trust through ML-infused proactive defense Analytics for NEXT-G and beyond ecosystems," *Procedia Comput. Sci.*, vol. 254, pp. 39–47, Feb. 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050925004120
[76] V. S. Krishna, J. Rairala, K. Raghuvardhan, R. S. Selvan, and D. A. Kanth, "Implementing machine learning techniques for the anticipation of maintenance requirements in naval assets," in *Proc. IEEE ICTES*, 2024, pp. 1–5.
[77] J. Lim, C. Park, and J. Paik, "Enhancing naval mine detection through augmented side scan sonar data and synthetic image generation for manned and autonomous teaming systems," in *Proc. Int. Conf. Electron. Inf. Commun. (ICEIC)*, 2025, pp. 1–4.
[78] S. Kim, J. Ku, E. Lee, U. Zaman, and K. Kim, "AI-driven conversational voice communication for maritime autonomous surface ships," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIIC)*, 2025, pp. 0408–0413.
[79] K. Blount, "Using artificial intelligence to prevent crime: Implications for due process and criminal justice," *AI Soc.*, vol. 39, no. 1, pp. 359–368, 2024.
[80] I. T. Adams, "Automation and artificial intelligence in police body-worn cameras: Experimental evidence of impact on perceptions of fairness among officers," *J. Criminal Justice*, vol. 97, no. 3, 2025, Art. no. 102373.
[81] H. Dubravova, J. Cap, K. Holubova, and L. Hribnak, "Artificial intelligence as an innovative element of support in policing," *Procedia Comput. Sci.*, vol. 237, pp. 237–244, May. 2024.
[82] B. R. Ardabili et al., "Understanding policy and technical aspects of AI-enabled smart video surveillance to address public safety," *Comput. Urban Sci.*, vol. 3, no. 1, p. 21, 2023.
[83] K. Razzaq and M. Shah, "Advancing cybersecurity through machine learning: A scientometric analysis of global research trends and influential contributions," *J. Cybersecurity Privacy*, vol. 5, no. 2, p. 12, 2025.
[84] R. Cecere and G. R. Greco, "Quantum technologies in defense and aerospace: A scientometric analysis of emerging trends and opportunities," in *Proc. SSRN*, 2022, Art. no. 5265429.
[85] D. O. Oyewola and E. G. Dada, "Exploring machine learning: A scientometrics approach using bibliometrix and VOSviewer," *SN Appl. Sci.*, vol. 4, no. 5, p. 143, 2022.
[86] S. Muthumari and S. Raja, "Bibliometric analysis of defence science journal during 2005–2014: A study based on Scopus database," *COLLNET J. Scientometrics Inf. Manag.*, vol. 10, no. 2, pp. 273–287, 2016.